Household Customer Segmentation Report

Table of Contents

1. INTRODUCTION 3
2. DATA 4
3. METHODS 5
3.1. Data Merging 5
3.2. Data Preprocessing 5
3.2.1. Feature Engineering: Creation of New Variables 5
3.2.2. Dropping Variables 6
3.2.3. Dropping Rows 6
3.2.4. Checking Outlier at Transaction Level 7
3.2.5. Aggregation to Household Level 7
3.2.6. Ordinal Encoding of Categorical Variables 8
3.2.7. Centered Log-Ratio (CLR) Transformation 8
3.2.8. Yeo-Johnson Transformation of Numerical Variables 8
3.2.9. Scaling with RobustScaler 8
3.2.10. Correlation Analysis 8
3.3. Clustering Algorithms 9
3.3.1. Clustering Algorithms Selection 9
3.3.2. Hyperparameters Selection 10
3.3.3. Clustering Models Selection 11
3.3.4. Clustering Model Validation Measures 11
4. RESULTS 12
4.1. Selection of the Best Model Among Six Clustering Algorithms for Each Variable Group 12
4.1.1. RFM Variable Group 12
4.1.2 Demographic Variable Group 15
4.2. Selection of the Best Model Across Variable Groups and the Combined All-Variables Model 18
4.3. Comparison of Model 1 and Model 7 With Other Models Using Various Combinations of Good Differentiating Variables 22
4.4. Advanced Clustering Algorithms (GMM & SOM) 25

4.4.1. RFM Variable Group 25
4.4.2. Demographic Variable Group 27
4.5. Principal Component Analysis (PCA) 28
4.6. Comparison of result with research article 29
4.6.1. RFM Variable 29
4.6.2. Demographic Variable 29
4.7. K-Means RFM (Model 1) Cluster Profile Analysis 30
4.7.1. RFM Variables 30
4.7.2. Product Preference Variables 33
4.8. K-Means Demographic (Model 7) Cluster Profile Analysis 35
4.8.1. Demographic Variables 35
4.8.2. RFM Variables 36
4.8.3. Other Behavioural Variables 37
4.8.4. Product Preference Variables 38
5. CONCLUSION 40
6. REFERENCES 41
7. APPENDIX 42

1. INTRODUCTION

Businesses often have a large number of customers with different demographics, purchasing behaviours and preferences. Treating all customers as one broad group leads to generic marketing strategies, while treating customers individually is impractical at scale, with both approaches causing inefficient marketing spending and lost opportunities for revenue. Customer segmentation strikes a balance by grouping customers into manageable segments based on common characteristics to identify valuable customers, understand purchasing patterns and deliver targeted marketing strategies. This project applied multiple unsupervised clustering algorithms to segment customers based on their shared characteristics. The algorithms included K-Means, Hierarchical Clustering with Complete, Average and Ward Linkages, Density-Based Spatial Clustering of Applications with Noise (DBSCAN), Hierarchical DBSCAN (HDBSCAN), Gaussian Mixture Model (GMM) and Self-Organizing Map (SOM). These algorithms were applied across multiple variable groupings, including RFM (Recency, Frequency, Monetary), demographic, behavioural, product preference and different combinations of these variables. Among these methods, two clustering models successfully identified meaningful customer clusters based on purchasing behaviour and demographic characteristics.

2. DATA

For this household customer segmentation project, we obtained the Complete Journey household-level transaction dataset from Kaggle website, provided by Dunnhumby (Gonen, 2020). A complete user guide with detailed information about this dataset can be obtained from GitHub (Github, 2015).

This household-level transaction dataset contains approximately 2.5 million transactions recorded over a two-year period from 2,500 households who regularly shop at the retailer. It serves as the primary dataset and is complemented by seven metadata datasets from the same sources. After merging and preprocessing, the combined dataset provides a detailed and multidimensional view of demographic information, household purchasing behaviour, product preferences and responses to marketing campaigns, which is valuable for household customer segmentation.

A brief description of the primary dataset and seven metadata datasets is summarized in Table 2-1, and a detailed description of each variable in the datasets can be found in Table 9-1 in Appendix.

No	Dataset	Dataset Types	Description
1	transaction	Primary	**• Lists every product purchased by households with each entry represents a line on a store receipt.** • Has 12 variables: household key, basket id, day, product id, quantity, sales value, store id, transaction time, week number, coupon match discount, coupon discount and retail discount.
2	demographic	Metadata	**• Contains demographic information for each household.** • Has 8 variables: household key, age description, marital status code, income description, homeowner description, household size description, household composition description and kid category description.
3	product	Metadata	**• Contains information on each product sold.** • Has 7 variables: product id, department, brand, commodity description, sub-commodity description, brand and size of product.
4	causal	Metadata	**• Indicates whether a product was featured in the weekly mailer or included in an in-store display.** • Has 5 variables: product id, store id, week number, mailer location and display location.
5	coupon	Metadata	**• Lists all the coupons sent to customers during a campaign and the products eligible for each coupon.** • Has 3 variables: coupon UPC, campaign and product ID.
6	coupon redemption	Metadata	**• Lists the coupons redeemed by each household.** • Has 4 variables: household key, day, coupon UPC and campaign
7	campaign	Metadata	**• Lists the campaigns received by each household.** • Has 3 variables: houshold key, campaign and campaign description.
8	campaign description	Metadata	**• Lists the duration of each campaign.** • Has 4 variables: campaign, campaign description, start day and end day.

Table 2-1. Brief description of the primary dataset and seven metadata dataset

3. METHODS

3.1. Data Merging

The primary household-level transaction datasets consisting of approximately 2.5 million rows was merged with seven metadata datasets using Dask (Dask, n.d.) because it processed data in chunks, so it does not need to load everything into memory at once that crashes Google Colab due to RAM limitations. Moreover, it split the data into smaller partitions and run operations in each partition in parallel across multiple CPU cores. Merging of these datasets was performed as shown in Table 3-1.

Step	Left Dataset	Right Dataset	Join Keys (Left Join)
1	Campaign dataset	Campaign dataset	campaign
2	Transaction dataset	Combined dataset (From Step 1)	household_key
3	Demographic dataset	Combined dataset (From Step 2)	household_key
4	Product dataset	Combined dataset (From Step 3)	product_id
5	Causal dataset	Combined dataset (From Step 4)	product_id, store_id, week_number
6	Coupon dataset	Coupon redemption dataset	coupon_upc, campaign
7	Combined dataset (From Step 5)	Coupon-enriched dataset (Step 6)	household_key, day, product_id, campaign

Table 3-1. Data merging steps and join keys.

3.2. Data Preprocessing

3.2.1. Feature Engineering: Creation of New Variables

We created several variables to capture temporal behaviour, promotional exposure, and purchasing intensity as shown in Table 3-2.

Variable Group	Variable Name	Data Type	Description	Construction Logic
Campaign & Promotion	received_ campaign	Binary (Yes/No)	Indicates whether a transaction occurred during an active campaign period	Set to **Yes** if transaction day falls within campaign start and end dates; otherwise **No**
Temporal	time_of_ day	Categorical	Time period in which the transaction occurred	Transaction timestamps categorized into **morning**, **afternoon**, **evening**, **night**, and **midnight**
Discount Usage	retail_ discount_ usage	Binary (Yes/No)	Indicates use of retail discounts	Set to **Yes** if retail discount amount > 0; otherwise **No**

Discount Usage	coupon_ discount_ usage	Binary (Yes/No)	Indicates use of coupon discounts	Set to **Yes** if coupon discount amount > 0; otherwise **No**
Product Categorization	new_ product_ category	Categorical	High-level product classification	Consolidates 44 granular product categories into **10 high-level categories** to reduce sparsity and improve interpretability
RFM Metrics	recency	Numerical	Days since last purchase	Calculated as number of days from last transaction date to dataset end date
RFM Metrics	frequency	Numerical	Transaction count	Total number of transactions made by a household during the observation period
RFM Metrics	monetary	Numerical	Household spending level	Total expenditure by a household during the observation period

Table 3-2. Description of newly created variables.

3.2.2. Dropping Variables

This project considered variables to be removed based on missingness, redundancy, or lack of segmentation relevance, as justified in Table 3-3 The project avoided to use imputation where missingness was excessive, because artificial value inflation could bias clustering outcomes.

Variable Group	Variables Dropped	Justification
Transactional	transaction time, campaign, start day, end day	Redundant after feature engineering
Coupon	coupon UPC, coupon discount	Aggregated into household-level features
Marketing placement	mailer location, display location	78.3% missing, no reliable imputation strategy
Demographics	marital status code, homeowner description	Excessive “Unknown” categories
Household composition	household composition description	Redundant and highly missing
Product identifiers	product ID, store ID, manufacturer, size, commodity description, sub-commodity description, quantity	Not meaningful for household segmentation

Table 3-3. Justification for excluding the variables.

3.2.3. Dropping Rows

Approximately 45% missingness was observed across all seven demographic variables. Row was deleted only where demographic information was entirely absent. We avoided using imputation where missingness was excessive because artificial value inflation could bias clustering outcomes.

3.2.4. Checking Outlier at Transaction Level

Although outliers are present in the retail discount and coupon match discount variables, no abnormal positive values are observed, for example: We will keep these outliers because the values are valid in the transactions and removing them would further reduce the total number of rows in the dataset.

3.2.5. Aggregation to Household Level

Since each household may have multiple transactions, we grouped the transactional-level data by household key into 33 household-level variables as shown in Table 3-4.

No	Variables	Data Type	Description	No	Variables	Data Type	Description
1	**Household Key**	Integer	Unique numeric identifier assigned to each household. (801 unique values within range 1-2,499)	17	**Coupon Discount Amount**	Float	Average coupon discount amount that each household received across all the transactions.
RFM Variables				18	**Coupon Discount Usage Proportion**	Float	Proportion of transactions where each household used a coupon discount.
2	**Recency**	Integer	Number of days since the houshold's last transaction. Lower values indicate more recent activity.	19	**Coupon Discount Non-Usage Proportion**	Float	Proportion of transactions where each household did not use a coupon discount.
3	**Frequency**	Integer	Total number of transactions made by the household within the period in the dataset. Higher values indicate the household shops more often.	20	**Campaign Non-Reach Proportion**	Float	Proportion of transactions where the household did not receive the campaign and any coupons.
4	**Monetary**	Float	Total amount of money a household has spent within the period in the dataset. Higher values indicates the household spends more money overall.	21	**Campaign Reach Proportion (16 Coupons)**	Float	Proportion of transactions where the household received the campaign and exactly 16 coupons that were selected based on the household's prior purchase behaviour.
Demographic Variables				22	**Campaign Reach Proportion (All Coupons)**	Float	Proportion of transactions where the household received the campaign and all available coupons pertaining to that campaign.
5	**Age Group**	String	Age of the household's primary shopper classified into 6 groups (“19–24” to “65+”).	**Product Preference Variables**			
6	**Income Group**	String	Annual household income classified into 12 groups ("Under 15K" to "250K+")	23	**National Brand Proportion**	Float	Proportion of transactions spent on national brands.
7	**Household Size Group**	String	Number of people in the household classified into 5 groups (“1” to “5+”).	24	**Private Brand Proportion**	Float	Proportion of transactions spent on private brands.
8	**Children Count Group**	String	Number of children in the household classified into 4 groups (“0” to “3+”).	25	**General Grocery Proportion**	Float	Proportion of transactions spent on general grocery products.
Behavioural Variables				26	**Health Personal Care Proportion**	Float	Proportion of transactions spent on health and personal care products.
9	**Morning Proportion**	Float	Proportion of transactions made in the morning (6am–11:59am),	27	**Produce Proportion**	Float	Proportion of transactions spent on fruits and vegetables.
10	**Afternoon Proportion**	Float	Proportion of transactions made in the afternoon (12pm–4:59pm).	28	**Meat Proportion**	Float	Proportion of transactions spent on meat products.
11	**Evening Proportion**	Float	Proportion of transactions made in the evening (5pm–8:59pm).	29	**Deli & Bakery Proportion**	Float	Proportion of transactions spent on deli and bakery products.
12	**Night Proportion**	Float	Proportion of transactions made at night (9pm–11:59pm).	30	**Miscellaneous & Services Proportion**	Float	Proportion of transactions spent on miscellaneous items and services.
13	**Midnight Proportion**	Float	Proportion of transactions made at midnight (12am–5:59am).	31	**Seafood Proportion**	Float	Proportion of transactions spent on seafood products.
14	**Retail Discount Amount**	Float	Average retail discount amount that each household received across all the transactions.	32	**Home & Lifestyle Proportion**	Float	Proportion of transactions spent on home and lifestyle products.
15	**Retail Discount Usage Proportion**	Float	Proportion of transactions where each household used a retail discount.	33	**Entertainment & Leisure Proportion**	Float	Proportion of transactions spent on entertainment and leisure product or services.
16	**Retail Discount Non-Usage Proportion**	Float	Proportion of transactions where each household did not use a retail discount.	34	**Alcohol Proportion**	Float	Proportion of transactions spent on alcoholic beverages.

Table 3-4. Description of the primary dataset and seven metadata datasets.

3.2.6. Ordinal Encoding of Categorical Variables

We applied ordinal encoding to the four demographic categorical variables (e.g. age group, income group, household size and children count group), as shown in Table 3-5, instead of one-hot encoding because these categories follow a natural progression that preserves meaningful information, while avoiding the sparsity and noise introduced by one-hot encoding.

Variable Name	Category Type	Category Range	Encoding Rationale	Benefit to Clustering
age_ group	Demographic	19–24 to 65+	Categories follow a natural age progression from younger to older households	Preserves ordinal information while reducing feature sparsity
income_ group	Demographic	Under 15K to 250K+	Income levels exhibit a monotonic increase in purchasing power	Maintains economic hierarchy and improves distance-based clustering
household _size_ group	Demographic	1 to 5+ members	Household size increases sequentially	Captures scale effects without generating multiple binary features
children_ count_ group	Demographic	0 to 3+ children	Number of children follows an inherent ordering	Retains family lifecycle information efficiently

Table 3-5. Application of ordinal encoding to demographic variables.

3.2.7. Centered Log-Ratio (CLR) Transformation

Centered Log-Ratio (CLR) transformation was applied to 24 compositional variables to address correlations arising from constant-sum constraints. The zero values were replaced with small values (10^{-6}), then the data were renormalized before transformation. CLR was selected over Additive Log-Ratio (ALR) transformation and Isometric Log-Ratio (ILR) transformation because of its symmetry, interpretability, and absence of a reference category requirement.

3.2.8. Yeo-Johnson Transformation of Numerical Variables

Five numerical variables (recency, frequency, monetary, retail discount mean, and coupon discount mean) showed right skewness and contained outliers. This study applied Yeo-Johnson transformation instead of logarithmic transformation because it reduces skewness more effectively than log transformation and handled outliers better.

3.2.9. Scaling with RobustScaler

We applied RobustScaler instead of StandardScaler to all features (excluding household key) because it uses medians and interquartile ranges, which prevents CLR-transformed values (extreme positive and negative log ratios) and numerical outliers from disproportionately influencing Euclidean distances during clustering.

3.2.10. Correlation Analysis

A full correlation heatmap is shown in Figure 9-1 in Appendix to check the correlation between all variables. In Figure 3-1, retail discount usage and retail discount non-usage proportion,

coupon discount usage and coupon discount non-usage proportion, and national and private proportion shows perfect correlation (|r| = 1) to each other. Therefore, we dropped retail discount non-usage proportion, coupon discount non-usage proportion and private proportion before clustering because they added zero new information but add noise and redundancy. Although household size group, children count size group, campaign reach proportion (16 coupons), campaign reach proportion (all coupons) and campaign non-reach proportion clustering algorithms shows high correlation (|r| > 0.8), we kept them for clustering algorithms because each variable may contain some unique variance that helps distinguish the clusters.

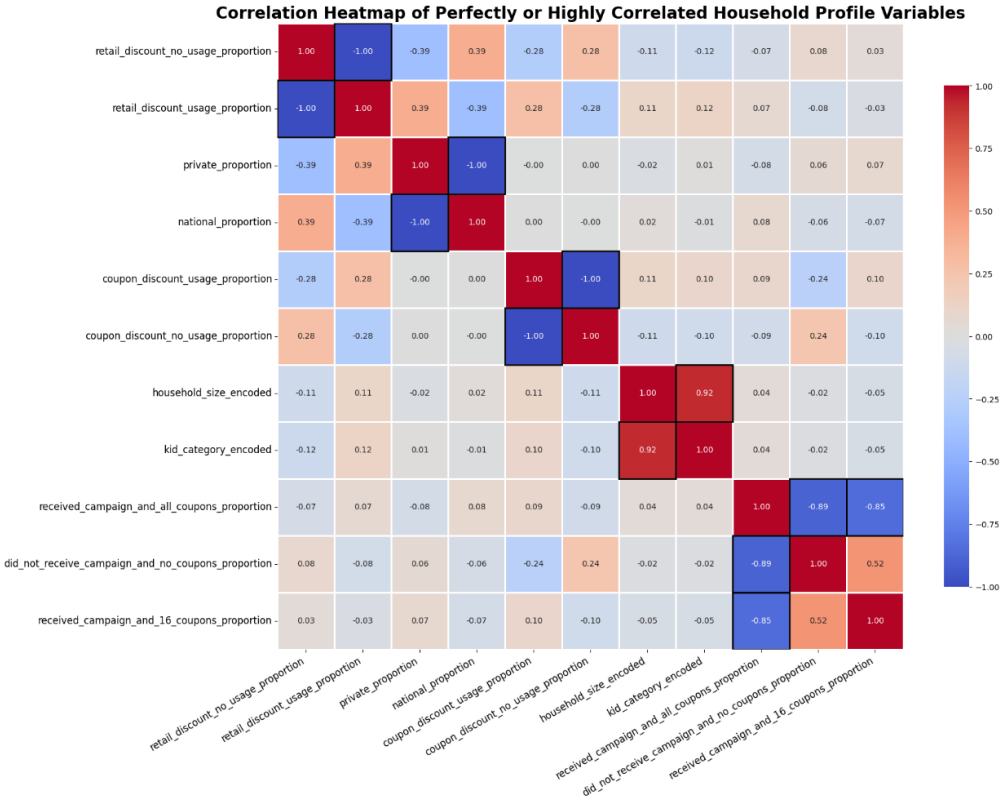


Figure 3-1. Correlation Heatmap of Perfectly or Highly Correlated Household Profile Variables.

3.3. Clustering Algorithms

3.3.1. Clustering Algorithms Selection

We used clustering algorithms, as described in Table 3-6, to identify distinct household segments based on purchasing behaviour, demographics, and product preference.

Clustering Family	Algorithm(s)	Core Assumption	Strengths	Role in This Study
Partition-based	K-Means	Clusters are approximately spherical and equally sized	Computationally efficient, highly interpretable, scalable to large datasets	Baseline and benchmark model for household segmentation
Hierarchical	Ward, Average, Complete linkage	Nested cluster structure exists	Does not require pre-specifying number of clusters, reveals	Explored cluster hierarchy and sensitivity to linkage criteria

			multi-level structure	
Density-based	DBSCAN, HDBSCAN	Clusters are dense regions separated by sparse areas	Identifies noise, captures irregular cluster shapes	Tested robustness to outliers and varying density patterns
Probabilistic	Gaussian Mixture Models (GMM)	Data generated from a mixture of probability distributions	Supports soft cluster membership, models overlapping clusters	Benchmarked probabilistic segmentation against hard clustering
Topology-preserving	Self-Organizing Maps (SOM)	Neighbourhood structure reflects similarity	Captures nonlinear relationships, preserves topology	Evaluated nonlinear segmentation structure in RFM and demographic space

Table 3-6. Comparison of clustering algorithm families.

3.3.2. Hyperparameters Selection

Table 3-7 below summarizes the hyperparameters used for each clustering algorithm.

Clustering Algorithm	Key Hyper-parameters	Approach	Evaluation Criteria	Outcome
K-Means	Number of clusters (*k*)	Determined using Elbow method, Silhouette scores, Davies–Bouldin index, Calinski-harabasz index	Compactness, separation, stability, business interpretability	Produced stable and interpretable clusters suitable for large-scale segmentation
Hierarchical Clustering	(Ward, Average, Complete)	Compared multiple linkage criteria to assess sensitivity to cluster compactness and separation	Cluster cohesion, dendrogram structure, interpretability	Provided insight into hierarchical relationships but limited scalability
DBSCAN	ε (epsilon), minPts	Iterative tuning across parameter ranges	Noise proportion, cluster density consistency	Classified a large proportion of households as noise due to high dimensionality
HDBSCAN	min_cluster_size, min_samples	Iterative density-based tuning	Cluster persistence, noise ratio	Sensitive to local density variation; limited practical segmentation utility
Gaussian Mixture Models (GMM)	Number of components, covariance structure	Tested multiple component counts and covariance types	Likelihood, cluster overlap, interpretability	Enabled probabilistic clustering and soft membership assignment

Self-Organizing Maps (SOM)	Grid size, learning rate	Selected to balance resolution and interpretability	Topological preservation, cluster readability	Captured nonlinear structure while maintaining interpretability

Table 3-7. Summary of clustering algorithms and their hyperparameter tuning strategies.

3.3.3. Clustering Models Selection

Each variable group (RFM, demographic, behavioural and product preference) was first tested using six different clustering algorithms: K-Means, Hierarchical Clustering (Ward Linkage, Average Linkage and Complete Linkage), DBSCAN and HDBSCAN, creating a total of 24 clustering algorithms. The results are discussed in results section 4.1.
The first 4 models were then selected as the best model chosen from each individual variables group after testing with 6 different clustering algorithms. A fifth model using K-Means on all 33 variables was also included. The results of these 5 models are compared and discussed in result section 4.2.

Next, the best model was then chosen used to compared with other models using K-Means algorithm and various combinations of good differentiating variables that were selected based on eta-squared and average Cohen's d (numerical variables or compositional variables) and Cramer's V and average cluster purity (categorical variables). The results of these models are compared and discussed in result section 4.3.

Additionally, we tested GMM and SOM on the RFM and demographic variables. The results of these models are compared and discussed in result section 4.4.

3.3.4. Clustering Model Validation Measures

Finally, the best clustering models were selected based on a combination of internal validation metrics (e.g. Silhouette score, Davies–Bouldin Index, Calinski-Harabasz Index), number of clusters, cluster distribution and business interpretability. Profile analysis was then performed on each cluster in the best clustering model for further business interpretation.

4. RESULTS

4.1. Selection of the Best Model Among Six Clustering Algorithms for Each Variable Group

In this subsection, each variable group (RFM, demographic, behavioural and product preference) was first tested using six different clustering algorithms: K-Means, Hierarchical Clustering (Ward Linkage, Average Linkage and Complete Linkage), DBSCAN and HDBSCAN, creating a total of 24 clustering algorithms. Their results are fully listed in Tables 9-2 and 9-3 in Appendix.

4.1.1. RFM Variable Group

Four different clustering validation measures are reported in this study, and their brief descriptions are provided in Table 4-1.

No	Validation Measures	Description
1	Silhouette score	It measures how well each data point fits within its assigned cluster compared to other clusters by considering cluster cohesion and separation (Artus, 2024). It ranges from -1 to 1, where higher values indicate that data points are well clustered within their assigned clusters and well separated from other clusters (Artus, 2024).
2	Davies-Bouldin Index (DBI)	It evaluates clustering quality by computing the ratio of cluster compactness (average distance between points and their cluster centroid) to cluster separation (distance between cluster centroids), where lower values indicate that data points are well clustered within their assigned clusters and well separated from other clusters (Artus, 2024).
3	Calinski-Harabasz Index (CHI)	It evaluates clustering quality by computing the ratio of between-cluster variance (sum of squared differences between cluster centroids and the overall centroid) to within-cluster variance (sum of squares differences between each point and its cluster centroid), where higher values indicate that data points are well clustered within their assigned clusters and well separated from other clusters (Moulavi et al., 2014).
4	Density-Based Clustering Validation Index (DBCV)	It is designed specifically for density-based clustering algorithms (DBSCAN and HDBSCAN) to evaluate clustering quality based on within-cluster density and between-cluster density (Chawla, 2023). It ranges from -1 to 1, where higher values indicate that high density within clusters and low density between clusters (Chawla, 2023).

Table 4-1. Description of clustering algorithm validation measured reported in this study.

In Table 4-2, Models 1, 5 and 6 had the highest Silhouette score (0.30), Model 3 had the lowest DBI (0.94) and Model 1 had the highest CHI (452.85). Based on these validation measures, Model 1 was selected as the best model because it achieved the highest scores in two of the three metrics.

Variables	Model No	Clustering Algorithms	Hyperparameter	Use PCA	Validation Measure				Number of Clusters	Cluster Proportion Description
					Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Index	DBCV Score		
1. Recency 2. Frequency 3. Monetary	1	K-Means	k = 4	No	0.30	1.03	452.85	-	4	The clusters are mildly imbalanced.
	2	Hierarchical Clustering (Ward Linkage)	k = 2		0.28	1.27	379.23	-	2	The clusters are mildly imbalanced.
	3	Hierarchical Clustering (Average Linkage)	k = 3		0.22	0.94	175.68	-	3	The clusters are moderately imbalanced.
	4	Hierarchical Clustering (Complete Linkage)	k = 2		0.26	1.01	218.46	-	2	The clusters are moderately imbalanced.
	5	DBSCAN	i) Epsilon = 0.4 ii) Minimum number of points within epsilon radius = 7		0.30	1.21	281.60	0.17	3	The clusters are midly imbalanced.
	6	HDBSCAN	i) Minimum number of points in a neigbourhood = 5 ii) Minimum number of points in a cluster = 20		0.30	1.17	267.96	0.27	3	The clusters are moderately imbalanced.

Table 4-2. Summary of the best clustering results for each clustering algorithm (K-Means, Hierarchical Clustering (Ward Linkage, Average Linkage and Complete Linkage), DBSCAN and HDBSCAN) using RFM variables.

There is no standard method to measure imbalance in multiclass classification. We adopted the binary classification approach, where the degree of imbalance is measured based on the minority class proportion: 0.20-0.40 is considered mild, 0.01-0.20 is moderate and less than 0.01 is severe imbalance (Mulugeta et al., 2023). Using this guideline and the cluster proportions listed in Table 9-3, the degree of imbalance for each RFM model was summarized in the cluster proportion description column in Table 4-2.

In Table 4-2, Models 1, 2 and 5 showed mildly imbalanced clusters, whereas Models 3, 4 and 6 showed moderately imbalanced clusters. Taking Models 1 and 3 as examples, Figure 4-1 shows that Model 1 had a minority class proportion of 0.23 and a relatively balanced household distribution. In contrast, Figure 4-2 shows that Model 3 had a minority class proportion of 0.01 and a highly imbalanced household distribution. Imbalanced clusters are not desirable because merging minority clusters into majority clusters can lead to overgeneralization of large clusters and loss of minority patterns (Lu et al., 2021). Therefore, Model 1 was preferred over Model 3 based on its more balanced cluster proportions.

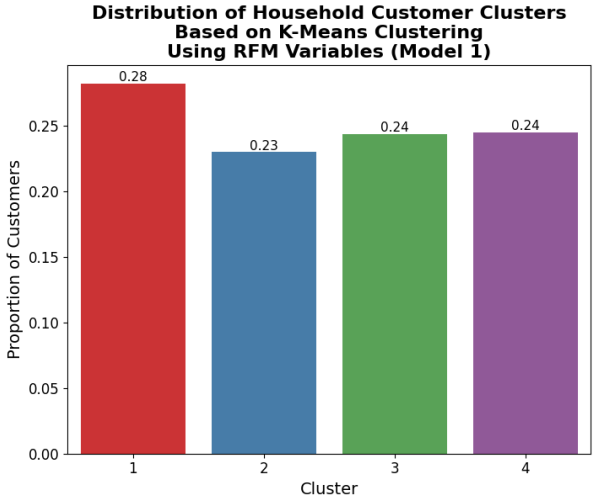


Figure 4-1. Model 1 (K-Means clusters) based on recency, frequency, and monetary variables.

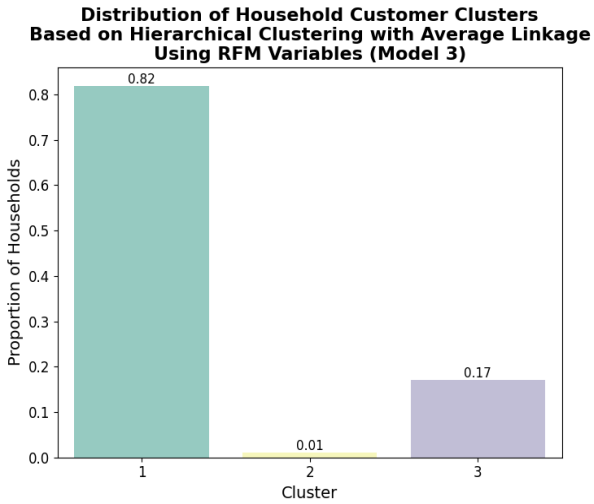


Figure 4-2. Model 3 (Hierarchical Clustering with Average Linkage cluster) based on recency, frequency, and monetary variables

We then examined the number of clusters and cluster profiles to assess whether the clusters were too few (oversimplifying household profiles) or too many (creating excessively detailed household profiles that complicate interpretation). Table 4-2 shows that only Model 1 produced four clusters, while other models produced two or three clusters. Using Models 1 and 5 as examples, we created RFM cluster profiles shown in Table 4-3 (Model 1) and Table 4-4 (Model 5) for analysis.

Table 4-3 shows that Model 1 produced four distinct and interpretable clusters with clear RFM value differentiation. The wider range of RFM values allowed the model to capture intermediate household segments between extreme clusters. For example, both Clusters 1 and 3 showed moderate RFM values between Clusters 2 and 4, allowing for more targeted marketing strategies for this group.

In Table 4-4, Model 5 produced only two simple clusters with a narrower gap in RFM values between Clusters 1 and 2 (Cluster 0 represented a noise group with no distinct pattern in DBSCAN and was excluded from the cluster profile analysis). Compared to Model 1, Model 5

was unable to identify intermediate segments and failed to capture the full diversity of household behaviour for effective marketing strategies.

RFM Variables (Model 1)					
No	Variables	Clusters (Median Per Cluster)			
		1	2	3	4
1	Recency	5	0	1	8
2	Frequency	158	279	122	80
3	Monetary	5816.68	9100.76	3194.17	2729.01

Table 4-3. Median values per cluster for RFM variables across the 4 clusters identified in Model 1 (K-Means).

RFM Variables (Model 5)				
No	Variables	Clusters (Median Per Cluster)		
		0	1	2
1	Recency	3	4	0
2	Frequency	121.5	131	188
3	Monetary	2439.94	4528.71	5946.97

Table 4-4. Median values per cluster for RFM variables across the 3 clusters identified in Model 5 (DBSCAN).

4.1.2 Demographic Variable Group

In this subsection, we examined the clustering results for the demographic variable group across six algorithms. As summarized in Table 4-5, Model 12 achieved the highest Silhouette score (0.56) and the lowest DBI (0.70), while Model 7 had the highest CHI (543.31). Based on these validation measures, Model 12 was selected as the best model because it achieved the highest scores in two of the three metrics.

Variables	Model No	Clustering Algorithms	Hyperparameter	Use PCA	Validation Measure				Number of Clusters	Cluster Proportion Description
					Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Index	DBCV Score		
1. Age Group 2. Income Group 3. Household Size Group 4. Children Count Group	7	K-Means	k = 4	Yes	0.38	0.90	543.31	-	4	The clusters are moderately imbalanced.
	8	Hierarchical Clustering (Ward Linkage)	k = 4		0.35	0.92	474.11	-	4	The clusters are moderately imbalanced.
	9	Hierarchical Clustering (Average Linkage)	k = 4		0.32	0.77	285.74	-	4	The clusters are severely imbalanced.
	10	Hierarchical Clustering (Complete Linkage)	k = 2		0.26	1.25	338.40	-	2	The clusters are mildly imbalanced.
	11	DBSCAN	i) Epsilon = 0.7 ii) Minimum number of points within epsilon radius = 7		0.41	1.12	273.89	0.68	17	The clusters are severely imbalanced.
	12	HDBSCAN	i) Minimum number of points in a neigbourhood = 5 ii) Minimum number of points in a cluster = 15		0.56	0.70	357.25	0.59	19	The clusters are severely imbalanced.

Table 4-5. Summary of the best clustering results for each algorithm (K-Means, Hierarchical Clustering (Ward Linkage, Average Linkage and Complete Linkage), DBSCAN and HDBSCAN) using demographic variables.

As shown in Table 4-5, Model 10 had a mildly imbalanced cluster, Models 7 and 8 had moderately imbalanced clusters, and Models 9, 11 and 12 had severely imbalanced clusters. Taking Models 7 and 12 as examples, Figure 4-3 shows that Model 7 had a minority class proportion of 0.157 and a moderately imbalanced household distribution. In contrast, Figure 4-4 shows that Model 12 had a minority class proportion of 0.006 and a highly imbalanced household distribution. Therefore, Model 7 was preferred over Model 12 based on its more balanced cluster proportions.

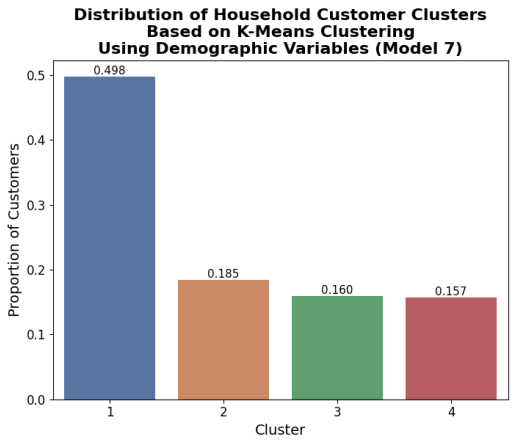


Figure 4-3. The proportion of households in each cluster for Model 7 (K-Means clusters) based on demographic variables.

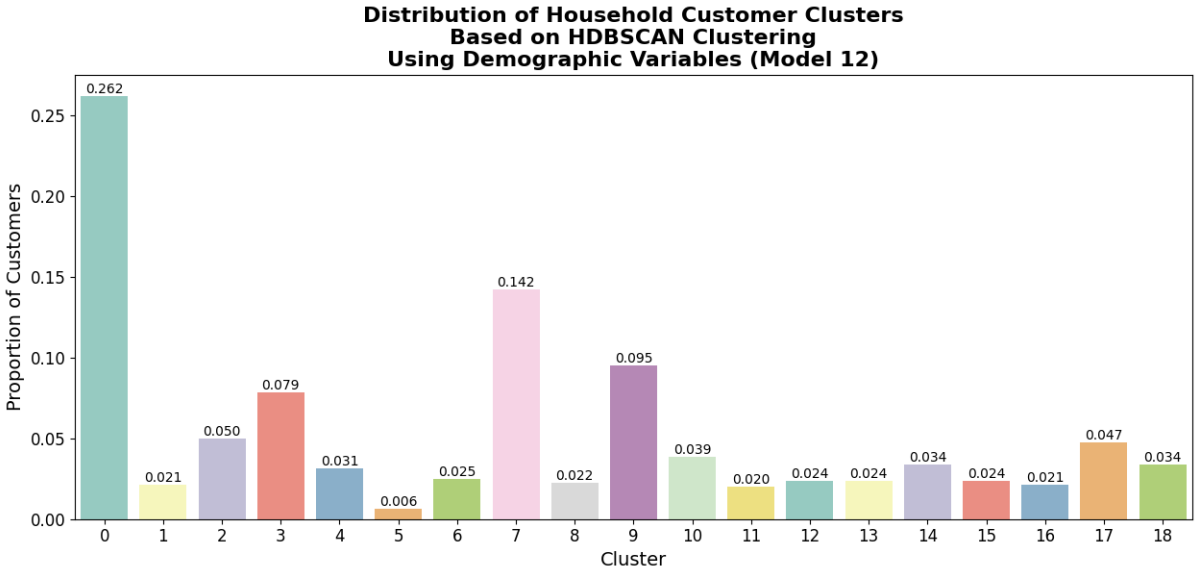


Figure 4-4. The proportion of households in each cluster for Model 12 (HDBSCAN clusters) based on demographic variables.

Next, we examined the number of clusters and cluster profiles to assess the interpretability for marketing purposes by creating demographic cluster profiles for Models 7, 10 and 12 as shown in Tables 4-6, 4-7 and 4-8 respectively. Table 4-6 shows that Model 7 produced four distinct and interpretable clusters. For example, Cluster 2 represented young adults (aged 25-34) with low income (35-49K), small household size (2 members) and no children (0), while Cluster 3 represented middle-aged adults (aged 35-44) with moderate income (50-74K), large household size (5+ members) with multiple children (3+).

In Table 4-7, Model 10 produced only two clusters with similar age profiles (Cluster 1: aged 35-44, Cluster 2: aged 45-54) and identical income levels (50-74K), household sizes (2 members) and no children (0). This limited segmentation did not capture meaningful household diversity. In contrast, Table 4-8 shows that Model 12 was over-segmented with eighteen clusters and one noise cluster (Cluster 0). Multiple clusters in Model 12 had similar profiles, for example, nine clusters shared the same age group (45-54), making business interpretation and profiling difficult. Therefore, Model 7 provided an optimal number of clusters with distinct profiles suitable for marketing strategies.

Demographic Variables (Model 7)					
No	Variables	Clusters (Mode Per Cluster)			
		1	2	3	4
1	Age Group	45-54	25-34	35-44	65+
2	Income Group	50-74K	35-49K	50-74K	35-49K
3	Household Size Group	2	2	5+	2
4	Children Count Group	0	0	3+	0

Table 4-6. Mode values per cluster for demographic variables across the four clusters identified in Model 7 (K-Means).

Demographic Variables (Model 10)			
No	Variables	Clusters (Mode Per Cluster)	
		1	2
1	Age Group	35-44	45-54
2	Income Group	50-74K	50-74K
3	Household Size Group	2	2
4	Children Count Group	0	0

Table 4-7. Mode values per cluster for demographic variables across the two clusters identified in Model 10 (Hierarchical Clustering with Complete Linkage

Demographic Variables (Model 12)											
No	Variables	Clusters (Mode Per Cluster)									
		0	1	2	3	4	5	6	7	8	9
1	Age Group	45-54	35-44	55-64	65+	19-24	45-54	35-44	35-44	25-34	25-34
2	Income Group	Under 15 K	35-49K	35-49K	35-49K	35-49K	75-99K	35-49K	50-74K	35-49K	50-74K
3	Household Size Group	3	5+	2	2	2	5+	3	2	3	1
4	Children Count Group	2	3+	0	0	0	3+	1	0	1	0
No	Variables	Clusters (Mode Per Cluster)									
		10	11	12	13	14	15	16	17	18	
1	Age Group	45-54	45-54	45-54	45-54	45-54	45-54	45-54	45-54	45-54	
2	Income Group	50-74K	Under 15 K	15-24K	25-34K	75-99K	35-49K	35-49K	50-74K	50-74K	
3	Household Size Group	3	1	1	2	2	1	2	1	2	
4	Children Count Group	1	0	0	0	0	0	0	0	0	

Table 4-8. Mode values per cluster for demographic variables across the nineteen clusters identified in Model 12 (HDBSCAN).

In summary, Model 1 and Model 7 were selected as the best models for the RFM and demographic variable groups respectively. Compared to the other five clustering algorithms, both K-Means models demonstrated strong validation measures, reasonably balanced cluster proportions and an optimal number of clusters that provided more interpretable and business-actionable segmentation. Only the model selection process and the results for RFM and demographic variable groups are discussed in this subsection as the behavioural and product preference variable groups follow the same process.

4.2. Selection of the Best Model Across Variable Groups and the Combined All-Variables Model

Four models (RFM K-Means Model 1, Demographic K-Means Model 7, Behavioural DBSCAN Model 17 and Product Preference DBSCAN Model 23) were selected as the best models after comparison with five other clustering algorithms within each individual variable group from the previous subsection 4.1. Their results are listed in Table 4-9. In addition, a RFM K-Means Model 25 that was trained on all 33 RFM, demographic, behavioural and product preference variables was added to Table 4-9 for comparison.

Variables	Model No	Clustering Algorithms	Validation Measure			Number of Clusters	Cluster Proportion Description
			Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Index		
RFM Variables 1. Recency 2. Frequency 3. Monetary	1	K-Means	0.30	1.03	452.85	4	The clusters are mildly imbalanced.
Demographic Variables 1. Age Group 2. Income Group 3. Household Size Group 4. Children Count Group	7	K-Means	0.38	0.90	543.31	4	The clusters are moderately imbalanced.
Behavioral Variables 1. Morning Proportion 2. Afternoon Proportion 3. Evening Proportion 4. Night Proportion 5. Midnight Proportion 6. Retail Discount Amount 7. Retail Discount Usage Proportion 8. Retail Discount Non-Usage Proportion 9. Coupon Discount Amount 10. Coupon Discount Usage Proportion 11. Coupon Discount Non-Usage Proportion 12. Campaign Non-Reach Proportion 13. Campaign Reach Proportion (16 Coupons) 14. Campaign Reach Proportion (All Coupons)	17	DBSCAN	0.44	0.92	256.37	3	The clusters are moderately imbalanced.
Product Preference Variables 1. National Brand Proportion 2. Private Brand Proportion 3. General Grocery Proportion 4. Health Personal CareProportion 5. Produce Proportion 6. Meat Proportion 7. Deli & Bakery Proportion 8. Miscellaneous & Services Proportion 9. Seafood Proportion 10. Home & Lifestyle Proportion 11. Entertainment & Leisure Proportion 12. Alcohol Proportion	23	DBSCAN	0.48	0.89	412.98	3	The clusters are moderately imbalanced.
All 33 Variables (Combined RFM, Demographic, Behavioural and Product Preference)	25	K-Means	0.17	1.90	99.69	4	The clusters are moderately imbalanced.

Table 4-9. Summary of the best clustering results across models for different variable groups (RFM, demographic, behavioral, product and combined variables).

Model 25 showed the lowest Silhouette score (0.17), the highest DBI (1.90) and the lowest CHI (99.69) among the five models in Table 4-9. Its poor performance was likely due to the inclusion of uncorrelated variables (e.g. afternoon proportion and entertainment & leisure proportion, |r| = 0.0) and weakly correlated variables (e.g. morning proportion and seafood proportion, |r| = 0.01), as shown in Figure 9-1 in the Appendix , which introduced random noise that masked useful information in the data (McCrory & Thomas, 2025). This observation aligned with the findings of McCrory et al. (2025), who demonstrated that increasing the number of irrelevant or uncorrelated variables led to lower Silhouette scores and higher DBI. Therefore, Model 25 was excluded from further analysis.

Based on Table 4-9, Model 23 had the highest Silhouette score (0.48) and the lowest DBI (0.89), and Model 17 had the second highest Silhouette score (0.44) and a comparatively low DBI (0.92). These metrics suggest both models perform well. However, Tables 4-10 & 4-11 show that there are no significant differences between the median values for most variables across clusters in Models 17 and 23 respectively. This limits the interpretability of the clusters and their usefulness for marketing purpose.

In Table 4-10, while some variables showed differences greater than 10% between clusters (e.g. morning, afternoon and evening proportions), the overall behavioural profiles of Clusters 1 and 2 in Model 17 were very similar (Cluster 0 represents noise and was excluded from this comparison). For example, campaign reach percentage (16 coupons) differed by only 0.26% (Cluster 1: 24.00%, Cluster 2: 24.26%), and retail discount amount differed by only $0.03 between the two clusters. Overall, this model failed to capture distinct behavioural profiles other than the three time-of-day variables.

Behavioural Variables (Model 17)				
No	Variables	Clusters (Median Per Cluster)		
		0	1	2
1	Morning Percentage	11.17%	26.09%	11.04%
2	Afternoon Percentage	36.00%	53.22%	38.59%
3	Evening Percentage	31.35%	12.95%	35.70%
4	Night Percentage	5.14%	0.00%	5.68%
5	Midnight Percentage	0.00%	0.00%	0.00%
6	Retail Discount Amount	$0.45	$0.48	$0.51
7	Discount Usage Percentage	45.88%	47.38%	50.24%
8	Discount Non-Usage Percentage	54.12%	52.62%	49.76%
9	Coupon Discount Amount	$0.00	$0.00	$0.00
10	Coupon Discount Usage Percentage	0.00%	1.42%	0.68%
11	Coupon Discount Non-Usage Percentage	100.00%	98.58%	99.32%
12	Campaign Non-Reach Percentage	77.83%	71.68%	65.17%
13	Campaign Reach Percentage (16 Coupons)	20.02%	24.00%	24.26%
14	Campaign Reach Percentage (All Coupons)	2.02%	5.33%	8.68%

Table 4-10.Median values per cluster for behavioural variables across the DBSCAN clusters identified in Model 17. (Note: Cluster 0 represents a noise group with no distinct pattern in DBSCAN.)

As shown in Table 4-11, none of the product preference variables demonstrated substantial differences between Clusters 1 and 2 in Model 23 (Cluster 0 represents noise and was excluded from this comparison). The largest difference between the two clusters was median produce percentage at only 2.67% (Cluster 1: 7.05%, Cluster 2: 9.72%). Even this largest difference was relatively small, indicating that these two clusters have nearly identical product preference profiles. This suggests that Model 23 failed to meaningfully segment households based on product preference variables.

Product Preference Variables (Model 23)				
No	Variables	Clusters (Median Per Cluster)		
		0	1	2
1	National Brand Percentage	73.23%	73.46%	72.24%
2	Private Brand Percentage	26.77%	26.54%	27.76%
3	General Grocery Percentage	61.26%	65.89%	63.66%
4	Health Personal Care Percentage	13.20%	12.46%	10.90%
5	Produce Percentage	6.82%	7.05%	9.72%
6	Meat Percentage	5.99%	6.18%	7.27%
7	Deli & Bakery Percentage	3.69%	3.47%	3.63%
8	Miscellaneous & Services Percentage	0.07%	1.26%	1.17%
9	Seafood Percentage	0.00%	0.00%	0.44%
10	Home & Lifestyle Percentage	0.08%	0.07%	0.12%
11	Entertainment & Leisure Percentage	0.00%	0.00%	0.00%
12	Alcohol Percentage	0.00%	0.00%	0.00%

Table 4-11. Median values per cluster for product preference variables across DBSCAN clusters identified in Model 23. (Note: Cluster 0 represents a noise group with no distinct pattern in DBSCAN.)

Furthermore, the household distribution in Figures 4-5 and 4-6 indicated that Models 17 and 23 had moderately imbalanced clusters with minority class proportions of 0.09 and 0.13 respectively. They exhibited greater imbalance in household distribution compared to Model 1 (Figure 4-1) and Model 7 (Figure 4-3).Therefore, Models 1 and 7 were preferred over Models 17 and 23 due to their more balanced cluster proportions.

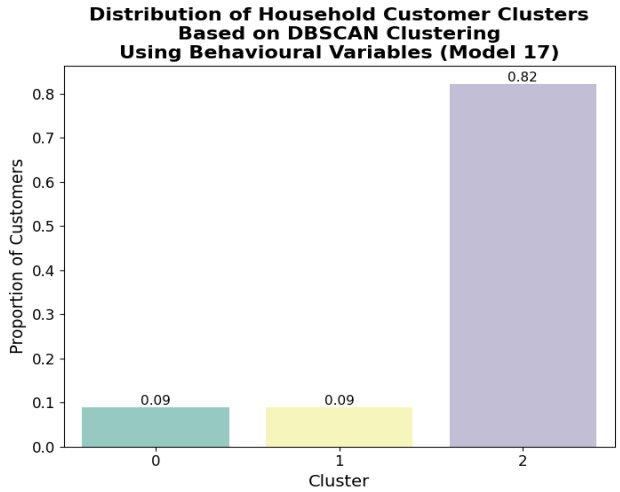


Figure 4-5. The proportion of households in each cluster for Model 17 (DBSCAN clusters based on behavioural variables).

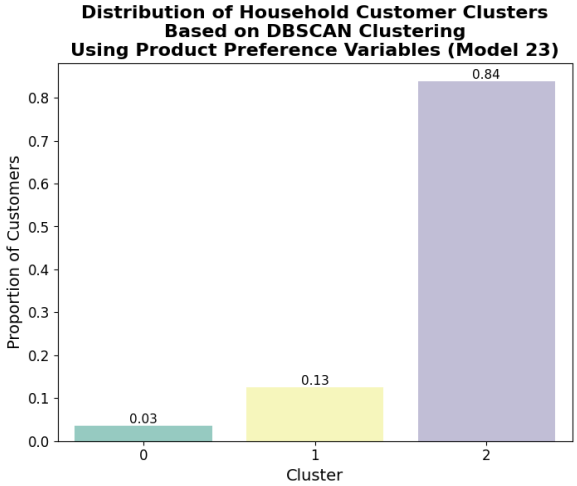


Figure 4-6. The proportion of households in each cluster for Model 23 (DBSCAN clusters based on product preference variables).

Models 1 and 7 had slightly lower Silhouette scores and slightly higher DBI than Models 17 and 23. However, they demonstrated better CHI, more balanced cluster proportions and more distinct cluster profiles that provided more interpretable and business-actionable segmentation. Thus, Models 1 and 7 were selected as the best models in this subsection.

Both Models 1 and 7 were selected to provide complementary insights about customer segments depending on specific business needs. Model 1 performed well at clustering RFM variables but poorly with demographic variables. In contrast, Model 7 clustered demographic variables well, but showed weak performance on RFM variables. This will be further discussed in Sections 4.7 and 4.8.

4.3. Comparison of Model 1 and Model 7 With Other Models Using Various Combinations of Good Differentiating Variables

In this subsection, Models 1 and 7 were compared with other models (Models 27-31) using various combinations of important differentiating variables. These variables were selected based on eta-squared and average Cohen's d for numerical variables, and Cramer's V and average cluster purity for categorical variables. The clustering results for these models are shown in Table 4-12.

Based on Table 4-12, running K-Means on different combinations of important differentiating variables showed little or no improvement. For example, after combining RFM variables with selected demographic and behavioral variables in Models 26 and 27, their validation measures worsened compared to Models 1 and 7. Similarly, Models 28-31 showed minimal or no improvement in their validation measures. This is likely due to the same issue discussed previously with Model 25, where irrelevant or weakly correlated variables introduced random noise that masked useful information in the data and weakened the distance relationships between the variables (McCrory & Thomas, 2025).

Variables	Model No	Clustering Algorithms	Validation Measure			Number of Clusters	Cluster Proportion Description
			Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Index		
RFM Segmentation 1. Recency 2. Frequency 3. Monetary	1	K-Means	0.30	1.03	452.85	4	The clusters are mildly imbalanced.
Demographic Segmentation 1. Age Group 2. Income Group 3. Household Size Group 4. Children Count Group	7	K-Means	0.38	0.90	543.31	4	The clusters are moderately imbalanced.
RFM, Demographic & Behavioral Variables 1. Recency 2. Frequency 3. Monetary 4. Age Group 5. Household Size Group 6. Children Count Group 7. Morning Proportion 8. Afternoon Proportion 9. Evening Proportion 10. Night Proportion	26	K-Means	0.22	1.49	240.52	4	The clusters are moderately imbalanced.
RFM & Demographic Varaibles 1. Recency 2. Frequency 3. Monetary 4. Age Group 5. Household Size Group 6. Children Count Group	27	K-Means	0.26	1.21	326.15	4	The clusters are moderately imbalanced.
RFM & DemographicVariables 1. Recency 2. Frequency 3. Monetary 4. Age Group	28	K-Means	0.30	1.12	387.18	4	The clusters are moderately imbalanced.
RFM & Demographic Variables 1. Recency 2. Frequency 3. Monetary 4. Household Size Group 5. Children Count Group	29	K-Means	0.33	1.05	532.00	3	The clusters are moderately imbalanced.
RFM & Behavioural Variables 1. Recency 2. Frequency 3. Monetary 4. Morning Proportion 5. Afternoon Proportion 6. Evening Proportion 7. Night Proportion	30	K-Means	0.30	1.30	409.08	3	The clusters are moderately imbalanced.
Demographic & Behavioural Variables 1. Age Group 2. Household Size Group 3. Children Count Group 4. Morning Proportion 5. Afternoon Proportion 6. Evening Proportion 7. Night Proportion	31	K-Means	0.30	1.17	362.94	4	The clusters are moderately imbalanced.

Table 4-12. Summary of the best clustering results across different K-Means models using different combinations of variables that are important for differentiating clusters.

This was further supported by the cluster profile analysis of Model 29 as shown in Table 4-13. In Model 29, the ranges for recency were 1-6, frequency was 97.5-210.5 and monetary value was $3068.34-$6599.52. These ranges were narrower compared to Model 1 with recency (1-8), frequency (80-279) and monetary ($2729.01-$9100.76). This reduction in RFM differentiation indicated that the inclusion of demographic variables (household size group and children count group) weakened the model's ability to distinguish distinct household profiles based on RFM. Models 26, 27, 28 and 30 were not discussed in this subsection as they showed the same issue of reduced differentiation as Model 29.

RFM, Demographic & BehaviouralVariables (Model 29)				
No	Variables	Clusters (Median or Mode Per Cluster)		
		1	2	3
1	Recency	1	2	6
2	Frequency	210.5	135	97.5
3	Monetary	6599.52	5915.73	3068.34
5	Household Size Group	2	5+	2
6	Children Count Group	0	3+	0

Table 4-13. Median values per cluster for selected RFM and behavioural variables across the four clusters identified in Model 29 (K-Means).

A similar situation was also observed in the cluster profile analysis of Model 31 as shown in Table 4-14, where there was a reduction in demographic differentiation after adding time-of-day variables (morning, afternoon, evening and night percentages). In Table 4-14, Clusters 2 and 4 shared the same age group (35-44) in Model 31. In contrast, Model 7 had four clusters with different age groups as shown in Table 4-6. This weakened demographic segmentation showed that adding irrelevant variables reduced the model's ability to create distinct demographic profiles.

Demographic & Behavioural Variables (Model 31)					
No	Variables	Clusters (Median or Mode Per Cluster)			
		1	2	3	4
1	Age Group	65+	35-44	45-54	35-44
2	Household Size Group	2	2	2	5+
3	Children Count Group	0	0	0	3+
4	Morning Percentage	24.41%	9.67%	11.08%	14.98%
5	Afternoon Percentage	52.83%	34.88%	39.48%	39.50%
6	Evening Percentage	11.90%	38.93%	33.05%	33.79%
7	Night Percentage	0.00%	6.72%	5.16%	5.46%

Table 4-14. Median or mode values per cluster for selected demographic and behavioural variables across the four clusters identified in Model 31 (K-Means).

In general, Models 1 and 7 had equal or higher Silhouette score, lower DBI and higher CHI compared to Models 26, 27, 28, 30 and 31. Although Model 29 had a slightly better Silhouette score and higher CHI than Model 1, Model 1 provided more distinct household profiles based on RFM variables that provided more interpretable clusters for marketing segmentation. As for Model 7, it performed better in all three validation measures than Model 29. Thus, Models 1 and 7 were still selected as the best models after the model selection process in this subsection.

4.4. Advanced Clustering Algorithms (GMM & SOM)

In this section, two additional advanced clustering algorithms (GMM & SOM) were tested on RFM and demographic variable groups to assess whether they could achieve better clustering results than Models 1 and 7.

4.4.1. RFM Variable Group

In Table 4-15, Models 32 had a slightly higher Silhouette score (0.31) and CHI (456.04) than Models 1 and 33. Based on these validation measures, Model 32 was selected as the best model because it achieved the highest scores in two of the three metrics. Model 32 was also preferred over Models 1 and 33 due to its more balanced cluster proportions.

Variables	Model No	Clustering Algorithms	Hyperparameter	Use PCA	Validation Measure			Number of Clusters	Cluster Proportion Description
					Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Index		
1. Recency 2. Frequency 3. Monetary	1	K-Means	k = 4	No	0.30	1.03	452.85	4	The clusters are mildly imbalanced.
	32	Gaussian Mixture Model	i) number of components = 2 ii) covariance type = full		0.31	1.20	456.04	2	The clusters are well-balanced.
	33	Self Organizing Maps	i) sigma = 1.0 ii) learning rate = 0.5 iii) k = 4		0.29	1.05	422.84	4	The clusters are mildly imbalanced.

Table 4-15.Summary of the best clustering results for each algorithm (K-Means, GMM and SOM) using RFM variables.

However, as shown in Table 4-16, Model 32 produced only two broad clusters: Cluster 1 (non-loyal and low-value households) and Cluster 2 (loyal and high-value households). There were no middle segments between these two clusters. In contrast, Model 1 in Table 4-3 produced finer segmentation with four clusters, each showing different meaningful RFM profiles that allow for more precise marketing decisions.

RFM Variables (Model 32)			
No	Variables	Clusters (Median Per Cluster)	
		1	2
1	Recency	5.5	1
2	Frequency	110.5	211
3	Monetary	3205.31	7090.07

Table 4-16.Median values per cluster for RFM variables across the 2 clusters identified in Model 32 (GMM).

As shown in Table 4-17, Model 33 produced four distinct and interpretable clusters with clear differentiation similar to Model 1 (Table 4-3). The two models were slightly different in how they segment households by monetary value. Model 33 had a larger gap of 6,960.99 (2,657.91 to 9,618.90) compared to Model 1's smaller gap of 6,371.75 (2,729.01 to 9100.76), suggesting Model 33 had better differentiation between the extreme low and high-value households. However, for mid-tier households, Model 33 had a smaller gap of 1,807.05 (3,712.14 to

5,519.19) compared to Model 1's larger gap of 2,622.51 (3,194.17 to 5,816.68), indicating that Model 1 had better differentiation between mid-tier households. Based on these trade-offs, the two models were tied in terms of monetary value differentiation.

To further distinguish between Models 1 and 33, we examined their validation measures and their cluster balance in Table 4-15. Comparing validation measures, Model 1 had slightly better performance with higher Silhouette score, lower DBI and higher CHI than Model 33. Both models had mildly imbalanced clusters. However, as shown in Figures 4-1 and 4-7, Model 1's cluster proportions (0.23 and 0.28) were more evenly distributed than Model 33's proportions (0.20 to 0.31).

RFM Variables (Model 33)					
No	Variables	Clusters (Median Per Cluster)			
		1	2	3	4
1	Recency	0	8	1	5
2	Frequency	133	80	310	153.5
3	Monetary	3712.14	2657.91	9618.9	5519.19

Table 4-17.Median values per cluster for RFM variables across the 4 clusters identified in Model 33 (SOM).

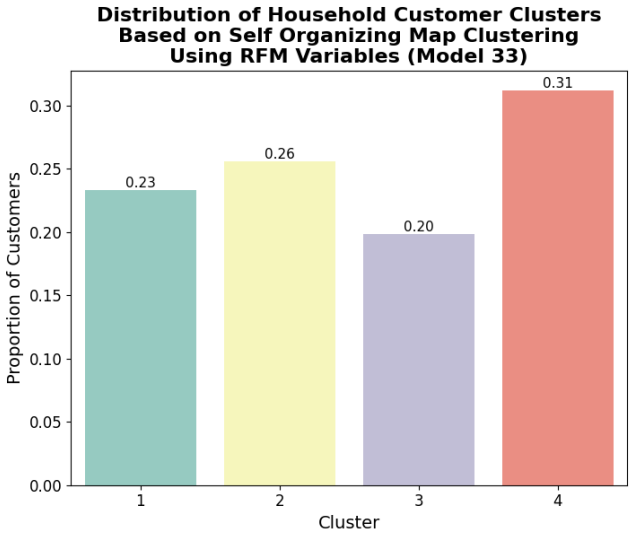


Figure 4-7. The proportion of households in each cluster for Model 33 (SOM clusters) based on RFM variables.

Compared to Model 32, Model 1 provided better cluster segmentation for targeted marketing strategies. Although both Models 1 and 33 produced four distinct and interpretable clusters, Model 1 demonstrated slightly better validation measures and more evenly distributed cluster proportions. Therefore, Model 1 was still selected as the best model compared with Models 32 and 33.

4.4.2. Demographic Variable Group

In Table 4-18, Model 7 had the highest Silhouette score (0.38), lowest DBI (0.90) and highest CHI (543.31). Based on these validation measures, Model 7 was selected as the best model because it achieved the highest scores across all three metrics.

Variables	Model No	Clustering Algorithms	Hyperparameter	Use PCA	Validation Measure			Number of Clusters	Cluster Proportion Description
					Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Index		
1. Age Group 2. Income Group 3. Household Size Group 4. Children Count Group	7	K-Means	k = 4	Yes	0.38	0.90	543.31	4	The clusters are moderately imbalanced.
	34	Gaussian Mixture Model	i) number of components = 4 ii) covariance type = full		0.35	1.01	468.16	4	The clusters are moderately imbalanced.
	35	Self Organizing Maps	i) sigma = 1.0 ii) learning rate = 0.5 iii) k = 3		0.36	0.92	476.38	3	The clusters are moderately imbalanced.

Table 4-18. Summary of the best clustering results for each algorithm (K-Means, GMM and SOM) using RFM variables.

As shown in Table 4-19, Model 34 showed less distinct household demographic profiles than Model 7 (Table 4-6), with Clusters 1 and 3 sharing the identical age group (35-44) and income group (50-74K). In contrast, Model 35 in Table 4-20 produced distinct household profiles similar to Model 7. However, Model 35 contained only three clusters and lacked an older adult cluster (aged 65+), which reduced the age range diversity needed for targeted marketing strategies.

Demographic Variables (Model 34)					
No	Variables	Clusters (Mode Per Cluster)			
		1	2	3	4
1	Age Group	35-44	45-54	35-44	65+
2	Income Group	50-74K	50-74K	50-74K	35-49K
3	Household Size Group	5+	2	2	2
4	Children Count Group	3+	0	0	0

Table 4-19. Mode values per cluster for demographic variables across the four clusters identified in Model 34 (GMM).

Demographic Variables (Model 35)				
No	Variables	Clusters (Mode Per Cluster)		
		1	2	3
1	Age Group	25-34	45-54	35-44
2	Income Group	35-49K	50-74K	50-74K
3	Household Size Group	1	2	5+
4	Children Count Group	0	0	3+

Table 4-20. Mode values per cluster for demographic variables across the four clusters identified in Model 35 (SOM).

Overall, Model 7 had better validation measures and distinctive household demographic profiles that covered a more diverse age range. Therefore, Model 7 was still selected as the best model compared with Models 34 and 35.

4.5. Principal Component Analysis (PCA)

Variables	Model No	Clustering Algorithms	Hyperparameter	Use PCA	Validation Measure		
					Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Index
1. Recency 2. Frequency 3. Monetary	1	K-Means	k = 4	No	0.30	1.03	452.85
				Yes	0.30	1.03	452.85
1. Age Group 2. Income Group 3. Household Size Group 4. Children Count Group	7	K-Means	k = 4	No	0.37	0.92	526.60
				Yes	0.38	0.90	543.31

Table 4-21. Comparison of clustering validation measures for Models 1 and 7 with and without PCA

We then compared the clustering validation measures for our best Models 1 and 7 with and without PCA as shown in Table 4-21. It was observed that the validation measures for K-Means RFM Model 1, with and without PCA, had the same validation scores (Silhouette score: 0.30, DBI: 1.03 and CHI: 452.85). This is most likely because RFM has only three variables in a low-dimensional space, so PCA does not improve the clustering results. For K-Means demographic Model 7, its validation scores became slightly better (Silhouette score: 0.38, DBI: 0.90 and CHI: 543.31) after applying PCA. This slight improvement is likely due to the four demographic variables (age, income, household size and children) were categorical variables that required ordinal encoding, which assigns arbitrary numerical values to groups in each variable that do not truly reflect Euclidean distances between groups. PCA created linear combinations of these encoded variables to capture the underlying structure that has more meaningful distance relationships than the ordinal encoded values.

4.6. Comparison of result with research article

4.6.1. RFM Variable

Study	Variables	Number of Clusters	Silhouette Score	Davies-Bouldin Index	Calinski–Harabasz Index
Our Study (K-Means RFM Model 1)	Recency, Frequency, Monetary	4	0.30	1.03	452.85
Abednego et al., 2023	Recency, Frequency, Monetary	5	0.30	1.19	NA
Ullah et al., 2023	Recency, Frequency, Monetary, Time	3	0.33	1.04	92.85

Table 4-22.Comparison of K-Means clustering validation measures for RFM-based segmentation with previous studies

Based on Table 4-22, we compared the validation measures of our K-Means RFM clustering study with two other similar studies using different datasets. All three studies showed comparable Silhouette scores, Davies-Bouldin Index values and number of clusters, indicating that our results are reasonable and consistent with existing research. Compared to Abednego et al. (2023), our study achieved the same Silhouette score (0.30) and a slightly better Davies-Bouldin Index (1.03). Compared to Ullah et al. (2023), although their study had better Silhouette score (0.33), our study outperformed theirs on both Davies-Bouldin Index (1.03) and Calinski-Harabasz Index (452.85). These differences may be caused by different methodologies. Both Abednego et al. (2023) and Ullah et al. (2023) assigned scores to their RFM variables, whereas we used the raw numerical RFM values. Additionally, Ullah et al. (2023) included an extra transaction time variable in their RFM clustering, which may have affected the result. Overall, our K-Means RFM Model 1 showed stronger cluster compactness and separation than these two other studies.

4.6.2. Demographic Variable

Study	Variables	Number of Clusters	Silhouette Score	Davies-Bouldin Index	Calinski–Harabasz Index
Our Study (K-Means Demographic Model 7)	Age Range, Income Range, Household Size, Children Count	4	0.38	0.90	543.31
Yulisasih et al., 2024	Sex, Marital Status, Age, Education Level, Income, Occupation, Settlement Size	4	0.43	NA	NA
Eva Yuliana et al., 2025	Age, Annual income, Spending Score	5	0.41	0.88	128.20

Table 4-23. Comparison of K-Means clustering validation measures for demographic-based segmentation with previous studies.

In Table 4-23, the validation measures of our K-Means demographic clustering study were compared with two other similar studies using different datasets. All three studies also showed comparable Silhouette scores, Davies-Bouldin Index values and number of clusters, indicating that our results are reasonable and consistent with existing research. Compared to our study, Yulisasih et al. (2024) had a better Silhouette score (0.43), while Yuliana et al. (2025) had a better Silhouette score (0.41) and Davies-Bouldin Index (0.88). These differences are likely due to their use of numerical values for age and annual income (the age and income in our study were categorical variables), and they included extra variables such as sex, marital status, education level, occupation, settlement size, and spending score. Therefore, Yulisasih et al. (2024) and Yuliana et al. (2025) showed stronger cluster compactness and separation than our K-Means demographic Model 7.

4.7. K-Means RFM (Model 1) Cluster Profile Analysis

4.7.1. RFM Variables

As Model 1 was one of our best models, we performed a detailed cluster profile analysis to better understand the characteristics of each cluster. Based on Table 4-3 and Figure 4-8, we were able to identify four household customer types from Model 1, which applied K-Means clustering algorithm to the RFM variables. The description of each cluster is summarized in Table 4-24.

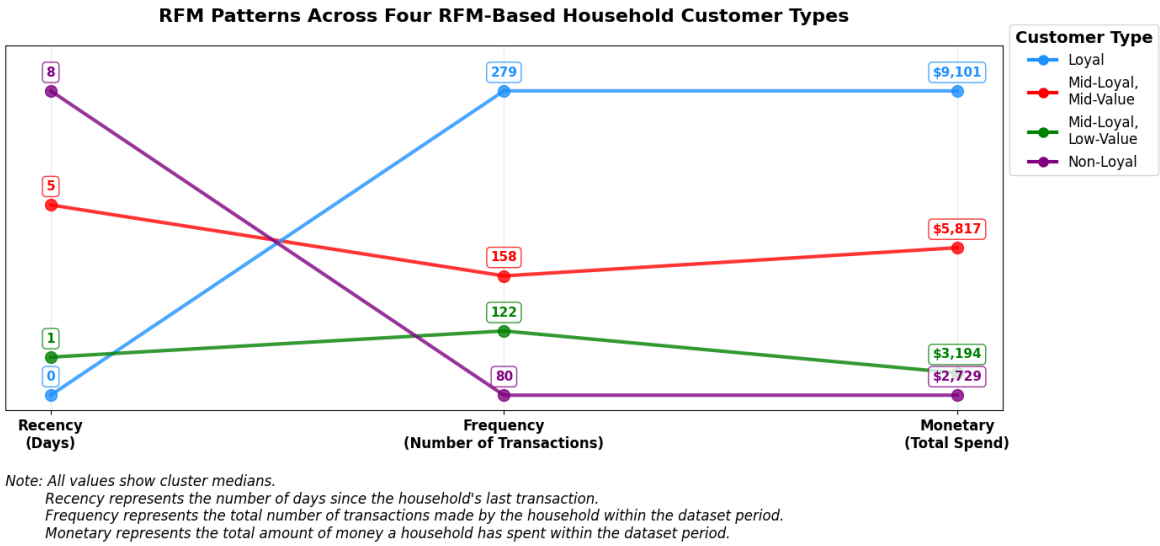


Figure 4-8. Parallel coordinates plot of median recency, frequency and monetary values across four RFM-based customer types identified using K-Means RFM clustering (Model 1).

Cluster No	Customer Type	Descriptions
1	Mid-Loyal, Mid-Value	• This cluster has moderate recency (5 days), moderate frequency (158 transactions) and moderate monetary value ($5,817). • It shows moderate engagement with better recency, frequency and monetary value than mid-loyal (low-value) cluster. • Mid-loyal (mid-value) and mid-loyal (low-value) clusters are differentiated by a large gap in monetary values which indicates a meaningful business distinction.
2	Loyal	• This cluster has the lowest recency (0 days), highest frequency (279 transactions) and highest monetary value ($9,101). • Its customers made a transaction today, bought most frequently and spent the most during the observed period in the dataset.
3	Mid-Loyal, Low-Value	• This cluster has low recency (1 day), moderate frequency (122 transactions) and low monetary value ($3,194). • It shows moderate engagement with lower recency, frequency and monetary value than mid-loyal (mid-value) cluster. • It consists of active customers who have not developed the purchasing frequency and spending levels compared to mid-loyal (mid-value) and loyal clusters.
4	Non-Loyal	• This cluster has the highest recency (8 days), lowest frequency (80 transactions) and lowest monetary value ($2,729). • Its customers made a transaction 8 days ago, bought infrequently and spent the least during the observed period in the dataset.

Table 4-24. Description of the RFM profiles for each cluster in Model 1 that applied K-Means clustering.

Overall, the four clusters form a clear hierarchy, where loyal customers represent our most engaged and valuable segment, followed by mid-loyal (mid-value) customers, then mid-loyal (low-value) customers, and finally the non-loyal customers as the least engaged and lowest value segment.

In Figure 4-9, the t-SNE visualisation of Model 1 shows different degrees of overlap and separation of the four RFM-based clusters using K-Means. Table 4-25 summarizes these degrees of overlapping and separation.

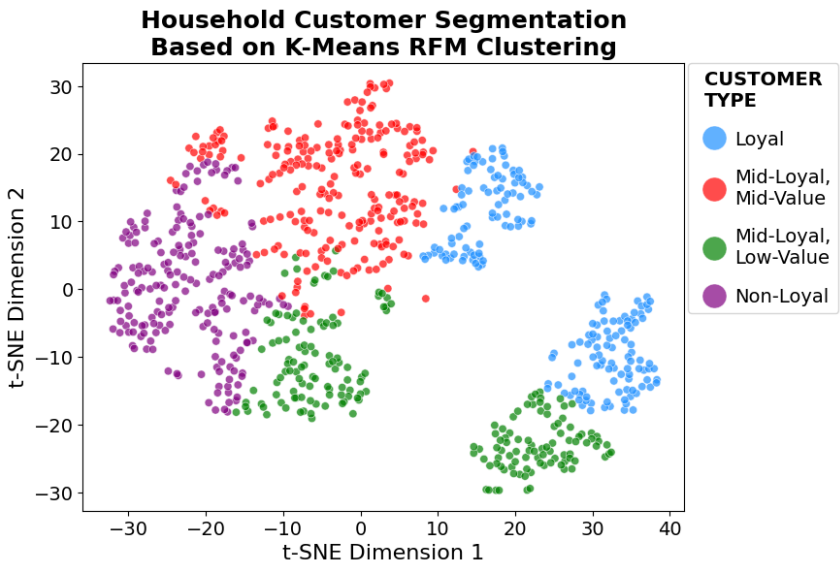


Figure 4-9. t-SNE visualisation of household customer segmentation based on K-Means RFM clustering using k=4 (Model 1).

Customer Type	Other Customer Type	Degree of Overlap/Separation Between Two Customer Types	Reason for Overlap / Separation
Mid-Loyal, Mid-Value	Loyal	Little Overlap	• Large gap in their recency values. • Moderate gap in their frequency and monetary values.
	Mid-Loyal, Low-Value	Most Overlap	• Small gap in their frequency values. • Moderate gap in their recency and monetary values.
	Non-Loyal	Medium Overlap	• Moderate gap in their recency, frequency and monetary values.
Loyal	Mid-Loyal, Low-Value	Little Overlap	• Large gap in their frequency and monetary values.
	Non-Loyal	Distinct Regions	• Very large gap in their recency, frequency and monetary values. • It reflects clear differentiation between high-engagement, high-value and low-engagement, low-value clusters.
Mid-Loyal, Low-Value	Non-Loyal	Little Overlap	• Very large gap in their recency values.

Table 4-25. Degree of overlap or separation between customer types in Model 1.

Overall, the t-SNE visualisation shows clear separation between loyal and non-loyal customer clusters due to their distinctive RFM values. In contrast, the mid-loyal (mid-value) and mid-loyal (low-value) customer clusters show the most overlap due to their similar RFM profiles.

4.7.2 Demographic Variables

As mentioned previously in Section 4.2, Model 1 had poor demographic differentiation across clusters. Table 4-26 shows that mid-loyal (mid-value), loyal and non-loyal clusters shared identical demographic profiles (age: 45-54, income: $50-74K, household size: 2, and children count: 0). The mid-loyal (low-value) cluster also had a similar profile, with the only difference being a lower income category ($35-49K). These similar demographic profiles suggested that Model 1 did not effectively segment customers based on demographic variables.

Demographic Variables (Model 1)					
No	Variables	Clusters (Mode Per Cluster)			
		Mid-Loyal, Mid-Value	Loyal	Mid-Loyal, Low-Value	Non-Loyal
1	Age Group	45-54	45-54	45-54	45-54
2	Income Group	50-74K	50-74K	35-49K	50-74K
3	Household Size Group	2	2	2	2
4	Children Count Group	0	0	0	0

Table 4-26. Mode values per cluster for demographic variables across the four clusters identified in Model 1 (K-Means).

4.7.3 Other Behavioural Variables

As shown in Figure 4-10, compared to other clusters, mid-loyal (mid-value) and loyal clusters had slightly higher percentages of morning transactions (13.17% and 12.90% respectively). Mid-loyal (low-value) cluster had a slightly higher percentage of night transactions (5.71%), while non-loyal cluster had slightly higher percentage of afternoon (39.80%) and evening (34.80%) transactions.

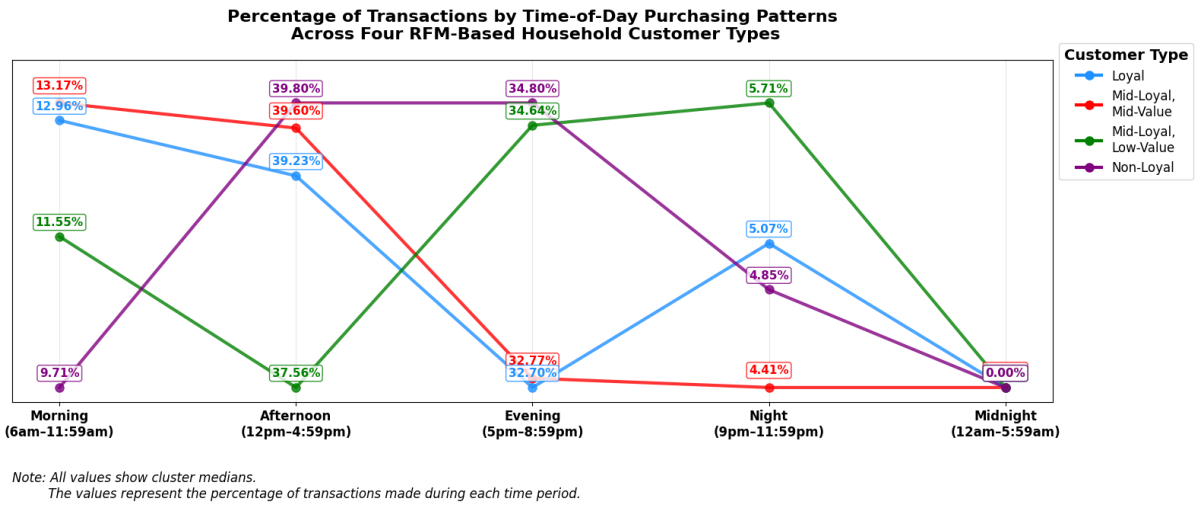
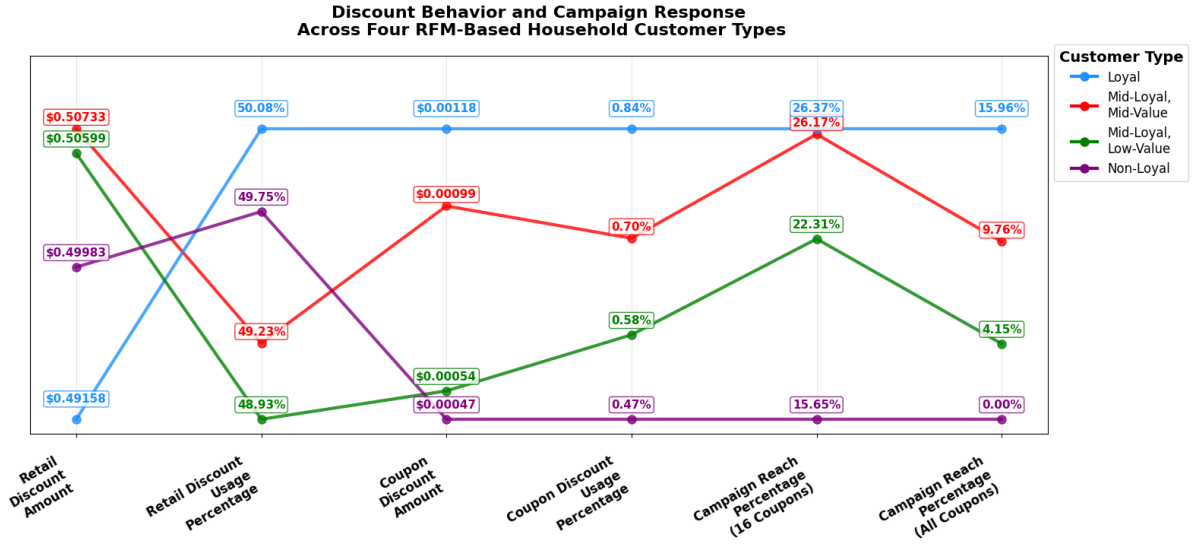


Figure 4-10. Parallel coordinates plot of median percentage of transactions by time-of-day purchasing patterns across four RFM-based customer types identified using K-Means RFM clustering (Model 1).

Figure 4-11 shows that the loyal cluster had the highest campaign coupon reach (16-coupon type: 26.37%, all-coupon type: 15.96%) and coupon discount usage (0.84%) compared to other clusters. This indicated that these loyal customers were not only targeted by promotional

campaigns but were also responsive to coupon discounts. In contrast, non-loyal cluster had the lowest campaign coupon reach (16-coupon type: 15.65% and all-coupon type: 0.00%) and coupon discount usage (0.47%), indicating that these non-loyal customers had minimal exposure to promotional campaigns and limited coupon redemption. Both mid-loyal (mid-value) and mid-loyal (low value) clusters showed moderate campaign coupon reach and coupon discount usage that was consistent with their RFM profiles of moderate level engagement and low-to-mid value.

When we analysed the average retail discounts in Figure 4-11, all clusters had similar average retail discount amounts per transaction (around $0.49-$0.50). However, they showed some slight differences in retail discount usage. The loyal cluster had the highest retail discount usage percentage (50.08%) compared to other clusters, suggesting these customers actively sought retail discounts in addition to coupon discounts. Both mid-loyal (low-value) and mid-loyal (mid-value) clusters had the lowest retail discount usage percentages (48.93% and 49.23% respectively) and moderate coupon usage, indicating these customers used a balanced mix of retail and discount coupons. Compared to other clusters, the non-loyal cluster had a moderate retail discount usage percentage (49.75%) despite having the lowest coupon discount usage (0.47%), suggesting they were opportunistic retail discount seekers who could potentially be converted to mid-loyal customers through targeted promotional campaigns.



Note: All values show cluster medians.
Discount amounts represent average discount per transaction.
Discount usage precentages represent the percentage of transactions with discounts applied.
Campaign reach percentage (16 coupons) represents the percentage of transactions where household received 16 personalized coupons.
Campaign reach percentage (All coupons) represents the percentage of transactions where household received all available coupons.

Figure 4-11.Parallel coordinates plot of median discount amounts and median percentage of transactions by discount usage and campaign reach across four RFM-based customer types identified using K-Means RFM clustering (Model 1).

4.7.2. Product Preference Variables

Based on Figure 4-12, all clusters demonstrated a clear preference for national brand products (70.92%-74.67%) over private brand products (25.33%-29.08%). Compared to other clusters, the mid-loyal (mid-value) cluster had the highest percentage of national brand product transactions (74.67%), while the mid-loyal (low-value) cluster had the highest percentage of private brand product transactions (29.08%). National brand products (e.g. Wattie's and Coca-Cola) are produced by well-known manufacturers and sold across multiple retailers

internationally. Private brands (e.g. Foodstuffs' Pams and Woolworths' Essentials) are owned and branded by the retailers and often manufactured by third-party suppliers.

Comparing the product preference patterns across clusters in Figure 4-12, the mid-loyal (mid-value) cluster allocated the highest percentage of transactions in the produce category (9.68%). The loyal cluster showed the highest percentages across multiple product categories, such as seafood (0.41%), healthcare & personal care (12.98%), miscellaneous & services (2.01%), home & lifestyle (0.17%) and deli & bakery (3.98%). The mid-loyal (low-value) cluster had the highest percentage in general grocery (64.66%), while non-loyal customers had the highest percentage in meat (7.99%).

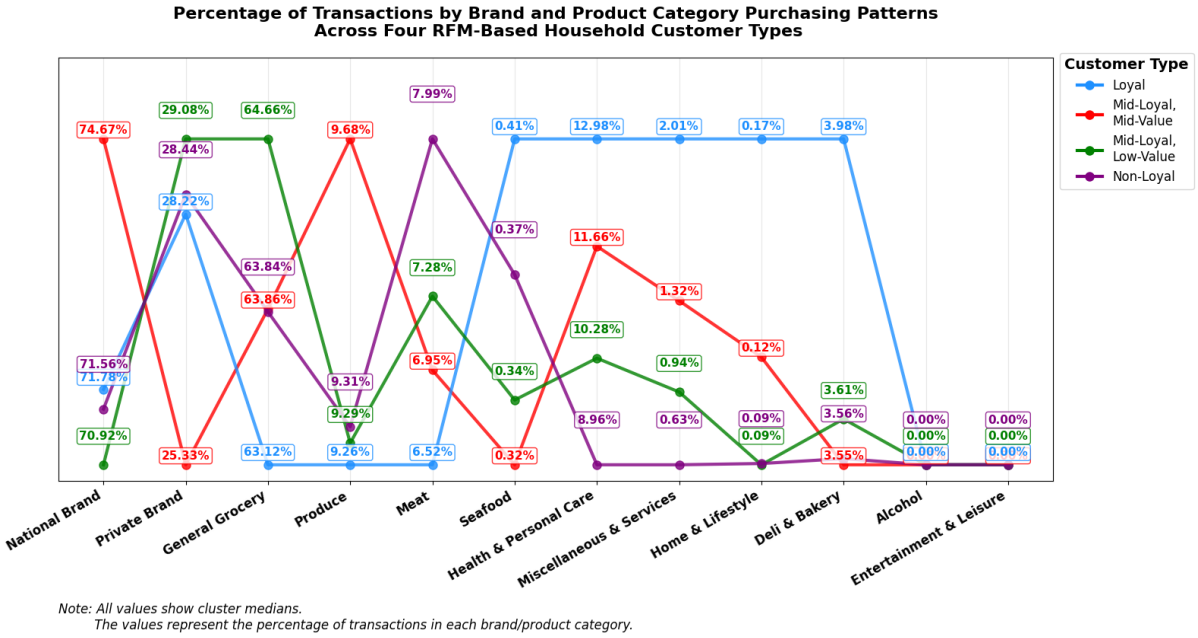


Figure 4-12. Parallel coordinates plot of median percentage of transactions by brand and product category purchasing patterns across four RFM-based customer types identified using K-Means RFM clustering (Model 1).

As the non-loyal customer cluster represents our at-risk customers, we selected it as an example for business and marketing recommendations. The behavioural and product preference patterns have been summarized in Table 4-27. Compared to other clusters, non-loyal customers tended to shop in the afternoon (39.80%) and evening (34.80%), and they had slightly higher spending on meat products (7.99%). This cluster also showed the lowest coupon exposure (15.65% for 16-coupon type and 0.00% for all-coupon type) and the lowest coupon redemption (0.47%). Based on this analysis, we recommend sending time-limited promotions and campaigns during afternoon and evening hours, particularly focused on meat products (for example, meat bundles with discounted sauces to encourage higher basket sizes).

Customer Type	Time-of-Day	Campaign Coupon Reach (16 Coupons)	Campaign Coupon Reach (All Coupons)	Coupon Discount Usage	Product Preference
Non-Loyal	• Afternoon (39.80%) • Evening (34.80%)	15.65%	0.00%	0.47%	Meat (7.99%)

Table 4-27. Summary of the percentage of transactions by time-of-day, campaign coupon reach, coupon discount usage and product preference for the non-loyal customer cluster.

4.8. K-Means Demographic (Model 7) Cluster Profile Analysis

4.8.1. Demographic Variables

As Model 7 was also one of our best models, we performed a detailed cluster profile analysis to better understand the characteristics of each cluster. Based on Table 4-6, we were able to identify four household customer types from Model 7, which applied K-Means clustering algorithm to the demographic variables. The description of each cluster is summarized in Table 4-28.

Cluster No	Customer Type	Descriptions
1	Middle-Aged Couple (Middle Income)	• This cluster is primarily made up of middle-aged adults (age: 45-54) with moderate incomes ($50-74K), small household sizes (2) and no children (0). • These customers are likely to have stable incomes with their children having moved out.
2	Young Couple (Low Income)	• This cluster is primarily made up of young adults (age: 25-34) with low incomes ($35-49K), small household sizes (2) and no children (0). • These customers are likely young couples who are early in their careers and have not yet started families.
3	Middle-Aged Large Family (Middle Income)	• This cluster is primarily made up of middle-aged adults (age: 35-44) with moderate incomes ($50-74K), large household sizes (5+) and multiple children (3+). • These customers share some characteristics with middle-aged couple (middle income), but they have large families.
4	Senior Couple (Low Income)	• This cluster is primarily made up of older adults (age: 65+) with low-to-moderate incomes ($35-49K), small household sizes (2) and no children (0). • These customers are likely retirees living in small households with their children having moved out.

Table 4-28. Description of the demographic profiles for each cluster in Model 7 that applied K-Means clustering.

Overall, the four customer clusters represent distinct houshold types based on age, family size and income level. They range from young couples with low incomes to senior couples with low income. These different demographic profiles influence their spending patterns.

In Figure 4-13, the t-SNE visualisation of model 7 shows four very distinct demographic-based clusters using K-Means. This clear separation occurs because the demographic variables are orginally categorical variables and were encoded into discrete values without intermediate values between categories

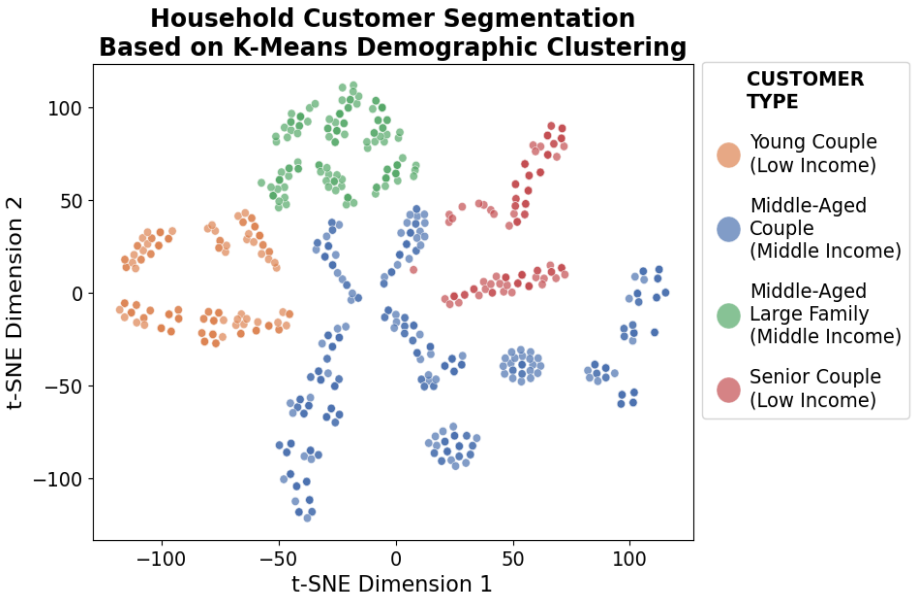


Figure 4-13. t-SNE visualisation of household customer segmentation based on K-Means demographic clustering using k=4 (Model 7).

4.8.2. RFM Variables

As mentioned previously in Section 4.2, Model 7 had poor RFM differentiation across clusters. Figure 4-14 shows that the recency was nearly identical across the four clusters with a narrow range of 2-3 days. Frequency was also very similar across the four clusters with a narrow range of 133-146 transactions. For monetary value, it showed some variations, but its range of $3,918 to $5,910 was still narrower compared to K-Means RFM Model 1. This poor RFM differentiation suggests that Model 7 does not effectively segment customers based on RFM dimensions.

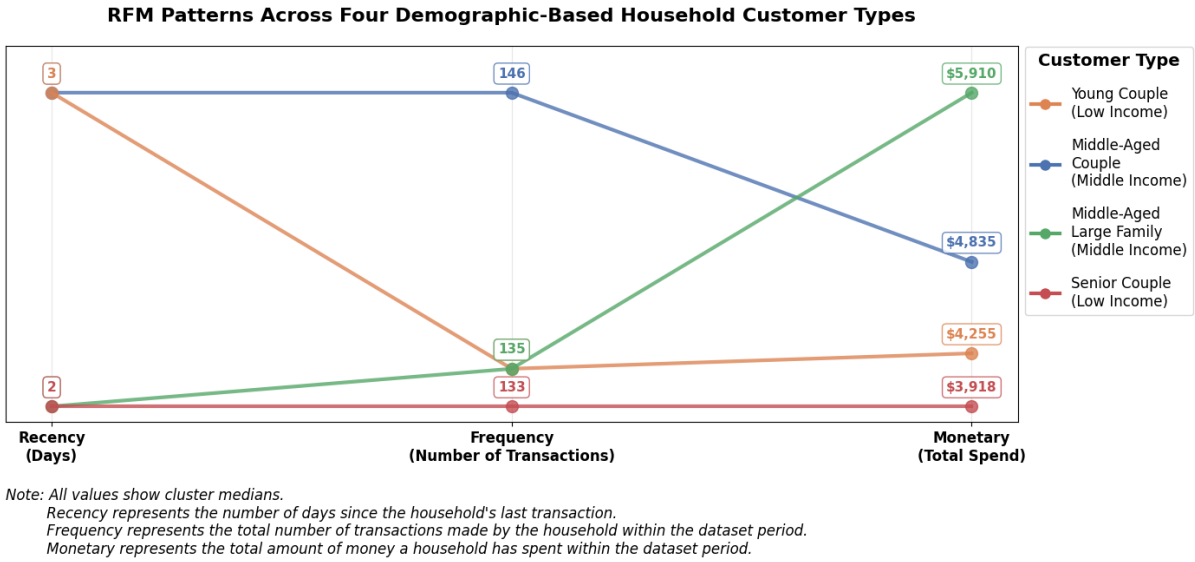


Figure 4-14. Parallel coordinates plot of median recency, frequency and monetary values across four demographic-based customer types identified using K-Means demographic clustering (Model 7).

4.8.3. Other Behavioural Variables

As shown in Figure 4-15, the young couple cluster had the highest percentages of evening (40.21%) and night (7.42%) transactions compared to other clusters, indicating that young adults tend to shop after traditional work hours. The middle-aged couple did not have the highest percentage in any time period, but it had the second-highest percentage of evening transactions (34.39%). The middle-aged large family cluster had the highest percentage of morning transactions (15.10%), suggesting that these customers are likely parents who shop after school drop-offs. Lastly, the senior couple cluster had the highest percentage of afternoon transactions (48.59%).

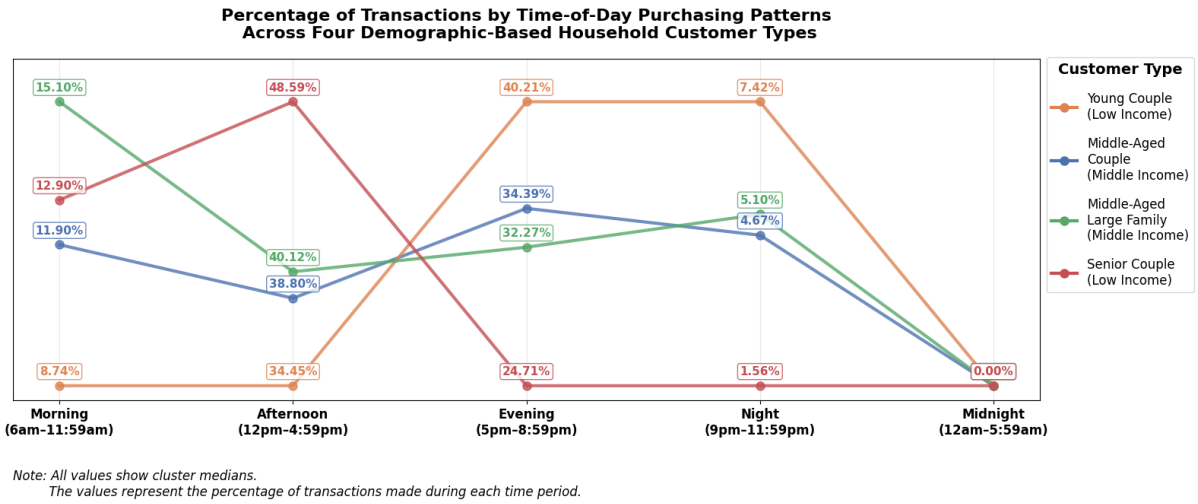
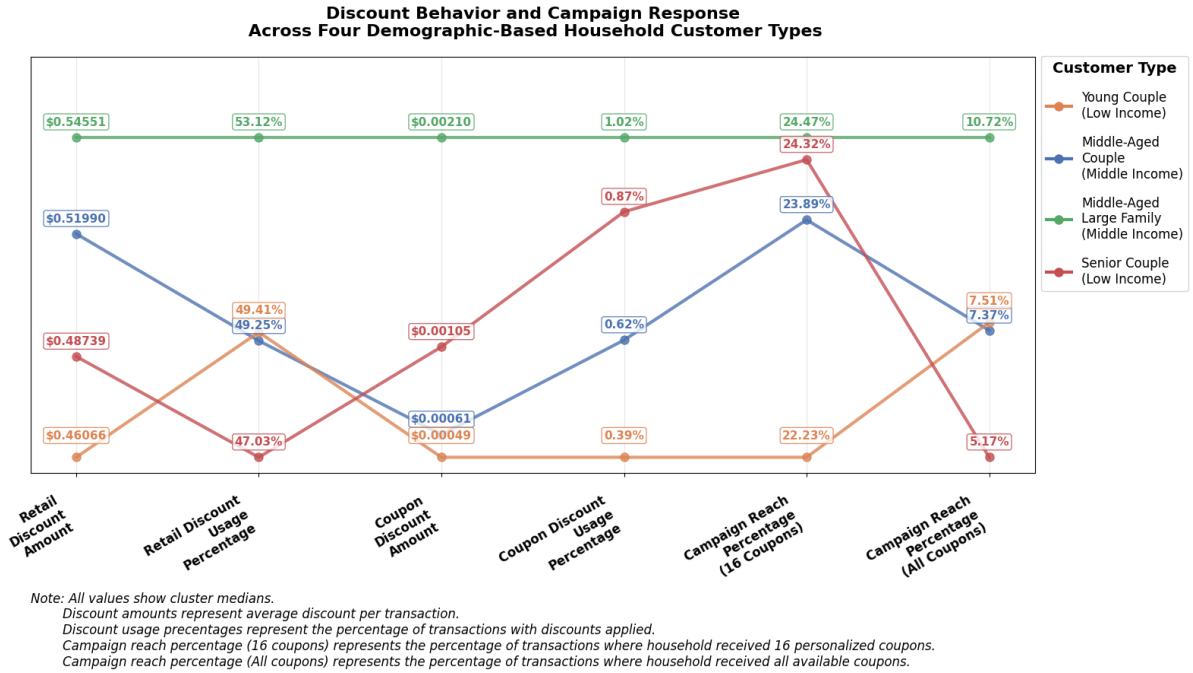


Figure 4-15. Parallel coordinates plot of median percentage of transactions by time-of-day purchasing patterns across four demographic-based customer types identified using K-Means demographic clustering (Model

Figure 4-16 shows that the middle-aged large family cluster had the highest total campaign coupon reach of 35.19% (16-coupon type: 24.47%, all-coupon type: 10.72%) and the highest coupon discount usage of 1.02%, indicating that they were heavily targeted by promotional campaigns and responsive to coupon discounts. The senior couple cluster had the lowest total coupon reach of 29.49% (16-coupon type: 24.32%, all-coupon type: 5.17%) but the second-highest coupon discount usage of 0.87%, suggesting that they had minimal campaign exposure but strong coupon redemption. The young couple cluster had the second-lowest total coupon reach of 29.74% (16-coupon type: 22.23%, all-coupon type: 7.51%) and the lowest coupon discount usage of 0.39%, indicating that they had minimal campaign exposure and low coupon redemption. Lastly, the middle-aged couple cluster showed moderate total campaign coupon reach of 31.26% (16-coupon type: 23.89%, all-coupon type: 7.37%) and moderate coupon discount usage of 0.62%.

In Figure 4-16, all clusters had similar average retail discount amounts per transaction (around $0.46-$0.55). However, they showed some differences in retail discount usage. The middle-aged large family cluster had the highest retail discount usage percentage (53.12%) compared to other clusters, suggesting these customers actively sought retail discounts in addition to coupon discounts, likely due to their larger family size and moderate incomes that increase budget sensitivity. In contrast, the senior couple cluster had the lowest retail discount usage percentage (47.03%). Both young couple and middle-aged couple clusters showed moderate retail discount usage percentages at 49.41% and 49.25% respectively.



Note: All values show cluster medians.
Discount amounts represent average discount per transaction.
Discount usage precentages represent the percentage of transactions with discounts applied.
Campaign reach percentage (16 coupons) represents the percentage of transactions where household received 16 personalized coupons.
Campaign reach percentage (All coupons) represents the percentage of transactions where household received all available coupons.

Figure 4-16. Parallel coordinates plot of median discount amounts and median percentage of transactions by discount usage and campaign reach across four demographic-based customer types identified using K-Means demographic clustering (Model 7).

4.8.4. Product Preference Variables

Based on Figure 4-17, all clusters demonstrated a clear preference for national brand products (72.05%-72.65%) over private brand products (27.35%-27.95%). Compared to other clusters, the middle-aged large family cluster had the highest percentage of national brand product transactions (72.65%), while the senior couple cluster had the highest percentage of private brand product transactions (27.95%).

Comparing the product preference patterns across clusters in Figure 4-17, the young couple cluster allocated the highest percentage of transactions to general grocery (65.50%) and health & personal care (11.65%). The middle-aged couple cluster had the highest percentage in home & lifestyle (0.13%). The middle-aged large family cluster did not have the highest percentage in any product category, but it had the second-highest percentage of transactions in meat (7.12%), general grocery (65.14%) and health & personal care (11.46%). The senior couple cluster showed the highest percentages across multiple product categories, such as produce (11.49%), meat (7.13%), deli & bakery (4.86%), miscellaneous & services (1.23%) and seafood (0.50%).

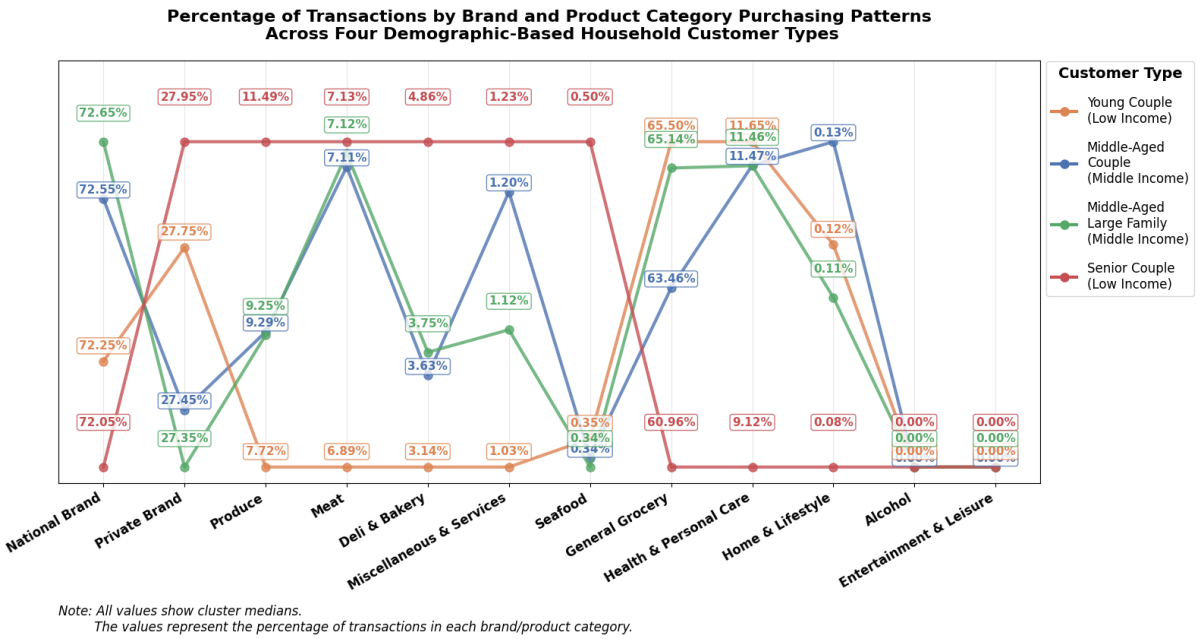


Figure 4-17. Parallel coordinates plot of median percentage of transactions by brand and product category purchasing patterns across four demographic-based customer types identified using K-Means demographic clustering (Model 7).

In this part, we selected the senior couple cluster as an example for business and marketing recommendations. The behavioural and product preference patterns have been summarized in Table 4-29. Compared to other clusters, the senior couple cluster tended to shop in the afternoon (48.59%), and they had slightly higher spending on produce (11.49%), meat (7.13%), deli & bakery (4.86%), miscellaneous & services (1.23%) and seafood (0.50%). This cluster also showed the lowest total coupon reach of 29.49% (16-coupon type: 24.32%, all-coupon type: 5.17%) but the second-highest coupon discount usage of 0.87%, suggesting that they had minimal campaign exposure but strong coupon redemption. Based on this analysis, we recommend distributing traditional physical promotions (such as printed coupons) during afternoon hours, particularly focused on their preferred product categories. This approach may be more effective for senior couple cluster because they may be less engaged with digital channels like app notifications and social media campaigns.

Customer Type	Time-of-Day	Campaign Coupon Reach (16 Coupons)	Campaign Coupon Reach (All Coupons)	Coupon Discount Usage	Product Preference
Senior Couple	Afternoon (48.59%)	24.32%	5.17%	0.87%	• produce (11.49%) • meat (7.13%) • deli & bakery (4.86%) • miscellaneous & services (1.23%) • seafood (0.50%)

Table 4-29. Summary of the percentage of transactions by time-of-day, campaign coupon reach, coupon discount usage and product preference for the senior couple.

5. CONCLUSION

We tested 35 different models using different clustering algorithms (e.g. K-Means, Hierarchical Clustering with Complete, Average and Ward Linkages, DBSCAN, HDBSCAN, GMM and SOM) with different variables (e.g. RFM, demographic, behavioural and product preference). The best models applied K-Means clustering algorithms to our RFM and demographic variables (Models 1 & 7). Each model produced 4 unique cluster profiles that are useful for business interpretation, as summarized in Table 5-1. These best models were selected based on their validation measures, the number of clusters, cluster distribution and business interpretability. Both models provide different insights about customers, and we can choose which one to use depending on business objectives.

Cluster No	Customer Type	
	K-Means RFM Model (Model 1)	**K-Means Demographic Model (Model 7)**
1	Mid-Loyal, Mid-Value	Middle-Aged Couple (Middle Income)
2	Loyal	Young Couple (Low Income)
3	Mid-Loyal, Low-Value	Middle-Aged Large Family (Middle Income)
4	Non-Loyal	Senior Couple (Low Income)

Table 5-1.Cluster profiles identified by K-Means RFM model (Model 1) and K-Means demographic model (Model 7).

6. REFERENCES

Dask. (n.d.). *Dask documentation.* https://doi.org/10.1002/0471721182

Github. (2015). C*SC465-visualization-project/Dataset/dunnhumby.* https://github.com/Lanbig/CSC465-visualization-project/blob/master/Dataset/dunnhumby%20-%20The%20Complete%20Journey%20User%20Guide.pdf

Gonen, F. (2020). *Dunnhumby - The Complete Journey*. Kaggle. Retrieved 1 December, 2025, from https://www.kaggle.com/datasets/frtgnn/dunnhumby-the-complete-journey/data?select=hh_demographic.csv.

Lu, Y., Cheung, Y. M., & Tang, Y. Y. (2021). Self-Adaptive Multiprototype-Based Competitive Learning Approach: A k-Means-Type Algorithm for Imbalanced Data Clustering. *IEEE Transactions on Cybernetics*, *51*(3). https://doi.org/10.1109/TCYB.2019.2916196

McCrory, M. N. F., & Thomas, S. A. (2025). Cluster Metric Sensitivity to Irrelevant Features. In *Computational Problems in Science and Engineering II*. https://doi.org/10.1007/978-3-031-78416-3-7

Mulugeta, G., Zewotir, T., Tegegne, A. S., Juhar, L. H., & Muleta, M. B. (2023). Classification of imbalanced data using machine learning algorithms to predict the risk of renal graft failures in Ethiopia. *BMC Medical Informatics and Decision Making*, *23*(1). https://doi.org/10.1186/s12911-023-02185-5

Yulisasih, B. N., Herman, H., Sunardi, S., & Yuliansyah, H. (2024). Evaluation of K-Means Clustering Using Silhouette Score Method on Customer Segmentation. *ILKOM Jurnal Ilmiah*, *16*(3). https://doi.org/10.33096/ilkom.v16i3.2325.330-342

7. APPENDIX

No	Table	Variable	Description
1	transaction_ data	HOUSEHOLD_KEY	Unique household identifier
2	transaction_ data	BASKET_ID	Unique transaction (basket) identifier
3	transaction_ data	DAY	Day on which the transaction occurred
4	transaction_ data	PRODUCT_ID	Unique product identifier
5	transaction_ data	QUANTITY	Number of units purchased
6	transaction_ data	SALES_VALUE	Total sales value of the transaction
7	transaction_ data	STORE_ID	Unique store identifier
8	transaction_ data	COUPON_MATCH_DISC	Retailer match of manufacturer coupon discount
9	transaction_ data	COUPON_DISC	Manufacturer coupon discount
10	transaction_ data	RETAIL_DISC	Retailer loyalty or promotional discount
11	transaction_ data	TRANS_TIME	Time of transaction
12	transaction_ data	WEEK_NO	Transaction week (1–102)
13	hh_demographic	HOUSEHOLD_KEY	unique household identifier
14	hh_demographic	AGE_DESC	estimated age range
15	hh_demographic	MARITAL_STATUS_CODE	marital status (a = married, b = single, u = unknown)
16	hh_demographic	INCOME_DESC	household income category
17	hh_demographic	HOMEOWNER_DESC	homeownership status (owner, renter, etc.)
18	hh_demographic	HH_COMP_DESC	household composition
19	hh_demographic	HOUSEHOLD_SIZE_DESC	household size (up to 5+)
20	hh_demographic	KID_CATEGORY_DESC	number of children present (up to 3+)
21	campaign_table	HOUSEHOLD_KEY	unique household identifier
22	campaign_table	CAMPAIGN	unique campaign identifier (1–30)
23	campaign_table	DESCRIPTION	campaign type (type a, type b, or type c)
24	campaign_desc	CAMPAIGN	unique campaign identifier (1–30)
25	campaign_desc	DESCRIPTION	campaign type (type a, type b, or type c)
26	campaign_desc	START_DAY	campaign start date
27	campaign_desc	END_DAY	campaign end date
28	product	PRODUCT_ID	unique product identifier
29	product	DEPARTMENT	product department category
30	product	COMMODITY_DESC	product commodity category
31	product	SUB_COMMODITY_DESC	product sub-commodity category

32	product	MANUFACTURER	manufacturer identifier
33	product	BRAND	brand type (private or national)
34	product	CURR_SIZE_OF_PRODUCT	package size (not available for all products)
35	coupon	CAMPAIGN	unique campaign identifier (1–30)
36	coupon	COUPON_UPC	unique coupon identifier (unique to household and campaign)
37	coupon	PRODUCT_ID	product identifier for which the coupon is redeemable
38	coupon_redempt	HOUSEHOLD_KEY	unique household identifier
39	coupon_redempt	DAY	transaction day
40	coupon_redempt	COUPON_UPC	unique coupon identifier (unique to household and campaign)
41	coupon_redempt	CAMPAIGN	unique campaign identifier
42	causal_data	PRODUCT_ID	unique product identifier
43	causal_data	STORE_ID	unique store identifier
44	causal_data	WEEK_NO	Transaction week (1–102)
45	causal_data	DISPLAY	in-store display location
46	causal_data	MAILER	mailer placement indicator

Table 7-1. Data dictionary for the original Complete Journey dataset

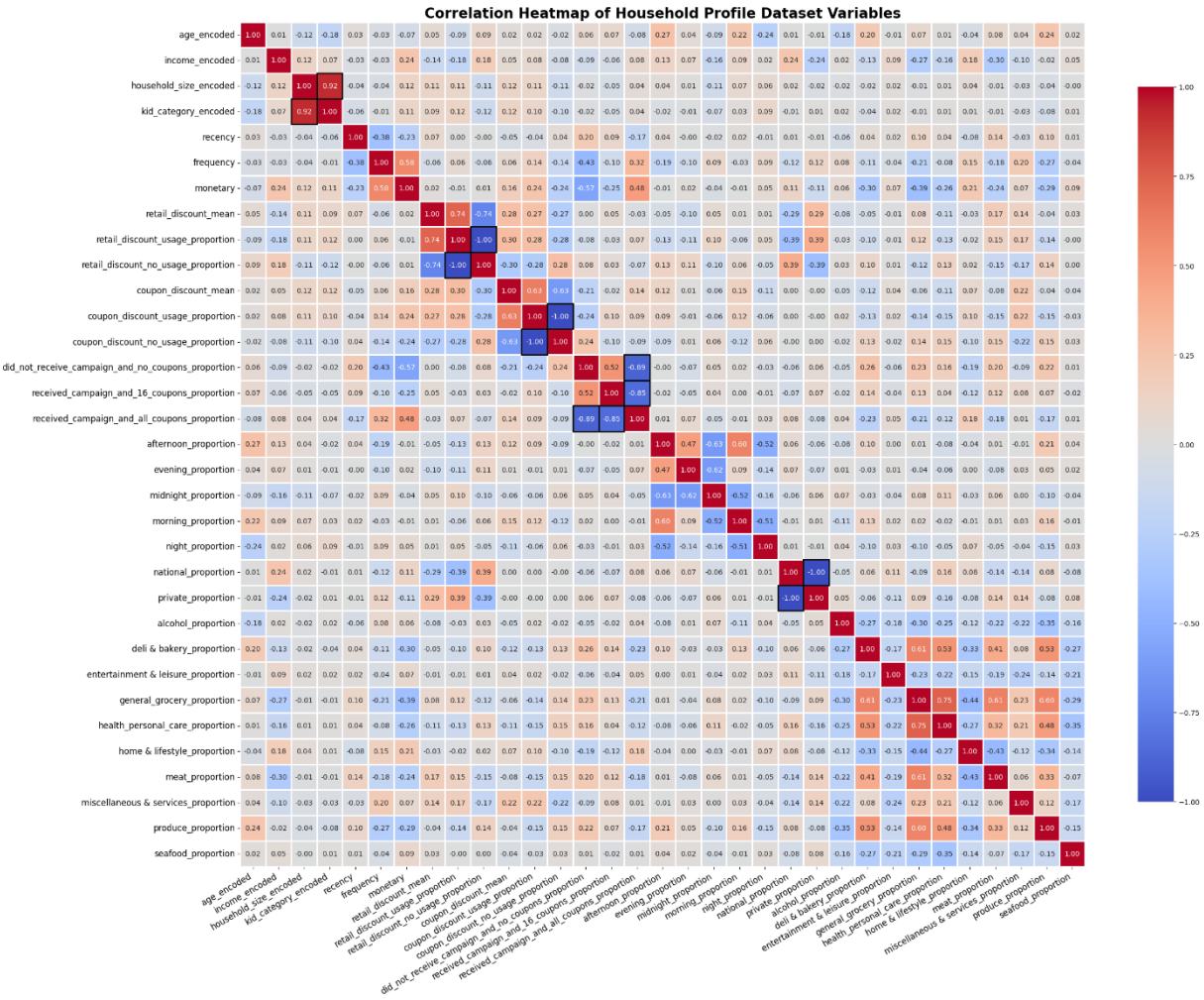


Figure 7-1. Correlation heatmap of 33 variables in household profile dataset.

Variables	Model No	Clustering Algorithms	Hyperparameter	Use PCA	Validation Measure			
					Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Index	DBCV Score
RFM Variables 1. Recency 2. Frequency 3. Monetary	1	K-Means	k = 4	No	0.30	1.03	452.85	-
	2	Hierarchical Clustering (Ward Linkage)	k = 2		0.28	1.27	379.23	-
	3	Hierarchical Clustering (Average Linkage)	k=3		0.22	0.94	175.68	-
	4	Hierarchical Clustering (Complete Linkage)	k=2		0.26	1.01	218.46	-
	5	DBSCAN	i) Epsilon = 0.4 ii) Minimum number of points within epsilon radius = 7		0.30	1.21	281.60	0.17
	6	HDBSCAN	i) Minimum number of points in a neigbourhood = 5 ii) Minimum number of points in a cluster = 20		0.30	1.17	267.96	0.27
Demographic Variables 1. Age Group 2. Income Group 3. Household Size Group 4. Children Count Group	7	K-Means	k = 4	Yes	0.38	0.90	543.31	-
	8	Hierarchical Clustering (Ward Linkage)	k = 4		0.35	0.92	474.11	-
	9	Hierarchical Clustering (Average Linkage)	k =4		0.32	0.77	285.74	-
	10	Hierarchical Clustering (Complete Linkage)	k =2		0.26	1.25	338.40	-
	11	DBSCAN	i) Epsilon = 0.7 ii) Minimum number of points within epsilon radius = 7		0.41	1.12	273.89	0.68
	12	HDBSCAN	i) Minimum number of points in a neigbourhood = 5 ii) Minimum number of points in a cluster = 15		0.56	0.70	357.25	0.59
Behavioral Variables 1. Morning Proportion 2. Afternoon Proportion 2. Evening Proportion 4. Night Proportion 5. Midnight Proportion 6. Retail Discount Amount 7. Retail Discount Usage Proportion 8. Retail Discount Non-Usage Proportion 9. Coupon Discount Amount 10. Coupon Discount Usage Proportion 11. Coupon Discount Non-Usage Proportion 12. Campaign Non-Reach Proportion 13. Campaign Reach Proportion (16 Coupons) 14. Campaign Reach Proportion (All Coupons)	13	K-Means	k=2	Yes	0.41	1.17	212.04	-
	14	Hierarchical Clustering (Ward Linkage)	k=2		0.39	1.08	192.62	-
	15	Hierarchical Clustering (Average Linkage)	k=2		0.51	0.92	59.83	-
	16	Hierarchical Clustering (Complete Linkage)	k=2		0.51	0.92	59.83	-
	17	DBSCAN	i) Epsilon = 2.4 ii) Minimum number of points within epsilon radius = 27		0.44	0.92	256.37	0.60
	18	HDBSCAN	i) Minimum number of points in a neigbourhood = 30 ii) Minimum number of points in a cluster = 10		0.44	0.91	246.22	0.60
Product Preference Variables 1. National Brand Proportion 2. Private Brand Proportion 3. General Grocery Proportion 4. Health Personal CareProportion 5. Produce Proportion 6. Meat Proportion 7. Deli & Bakery Proportion 8. Miscellaneous & Services Proportion 9. Seafood Proportion 10. Home & Lifestyle Proportion 11. Entertainment & Leisure Proportion 12. Alcohol Proportion	19	K-Means	k=2	Yes	0.46	1.00	380.02	-
	20	Hierarchical Clustering (Ward Linkage)	k=2		0.45	0.96	353.28	-
	21	Hierarchical Clustering (Average Linkage)	k=2		0.59	0.64	24.45	-
	22	Hierarchical Clustering (Complete Linkage)	k=2		0.59	0.64	24.45	-
	23	DBSCAN	i) Epsilon = 2.0 ii) Minimum number of points within epsilon radius = 23		0.48	0.89	412.98	0.68
	24	HDBSCAN	i) Minimum number of points in a neigbourhood = 5 ii) Minimum number of points in a cluster = 10		0.47	0.91	427.21	0.68

Table 7-2. Detailed information on validation measures, hyperparameters and usage of PCA for 35 clustering algorithms involved in the model selection

Variables	Model No	Clustering Algorithms	Hyperparameter	Use PCA	Validation Measure			
					Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Index	DBCV Score
RFM, Demographic, Behavioral & Product Preference Variables 1. Recency 2. Frequency 3. Monetary 4. Age Group 5. Income Group 6. Household Size Group 7. Children Count Group 8. Morning Proportion 9. Afternoon Proportion 10. Evening Proportion 11. Night Proportion 12. Midnight Proportion 13. Retail Discount Amount 14. Retail Discount Usage Proportion 15. Retail Discount Non-Usage Proportion 16. Coupon Discount Amount 17. Coupon Discount Usage Proportion 18. Coupon Discount Non-Usage Proportion 19. Campaign Non-Reach Proportion 20. Campaign Reach Proportion (16 Coupons) 21. Campaign Reach Proportion (All Coupons) 22. National Brand Proportion 23. Private Brand Proportion 24. General Grocery Proportion 25. Health Personal CareProportion 26. Produce Proportion 27. Meat Proportion 28. Deli & Bakery Proportion 29. Miscellaneous & Services Proportion 30. Seafood Proportion 31. Home & Lifestyle Proportion 32. Entertainment & Leisure Proportion 33. Alcohol Proportion	25	K-Means	k=4	Yes	0.17	1.90	99.69	-
RFM, Demographic & Behavioral Variables 1. Recency 2. Frequency 3. Monetary 4. Age Group 5. Household Size Group 6. Children Count Group 7. Morning Proportion 8. Afternoon Proportion 9. Evening Proportion 10. Night Proportion	26	K-Means	k=4	Yes	0.22	1.49	240.52	-
RFM & Demographic Variables 1. Recency 2. Frequency 3. Monetary 4. Age Group 5. Household Size Group 6. Children Count Group	27	K-Means	k=4	Yes	0.26	1.21	326.15	-
RFM & Demographic Variables 1. Recency 2. Frequency 3. Monetary 4. Age Group	28	K-Means	k=4	Yes	0.30	1.12	387.18	-
RFM & Demographic Variables 1. Recency 2. Frequency 3. Monetary 4. Household Size Group 5. Children Count Group	29	K-Means	k=3	Yes	0.33	1.05	532.00	-
RFM & Behavioural Variables 1. Recency 2. Frequency 3. Monetary 4. Morning Proportion 5. Afternoon Proportion 6. Evening Proportion 7. Night Proportion	30	K-Means	k=3	Yes	0.30	1.30	409.08	-
Demographic & Behavioural Variables 1. Age Group 2. Household Size Group 3. Children Count Group 4. Morning Proportion 5. Afternoon Proportion 6. Evening Proportion 7. Night Proportion	31	K-Means	k=4	Yes	0.30	1.17	362.94	-

Table 7-2 (continued). Detailed information on validation measures, hyperparameters and usage of PCA for 35 clustering algorithms involved in the model selection

Variables	Model No	Clustering Algorithms	Hyperparameter	Use PCA	Validation Measure			
					Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Index	DBCV Score
1. Recency 2. Frequency 3. Monetary	32	Gaussian Mixture Model	i) number of components = 2 ii) covariance type = full	No	0.31	1.20	456.04	-
	33	Self Organizing Maps	i) sigma = 1.0 ii) learning rate = 0.5 iii) k = 4		0.29	1.05	422.84	-
1. Age Group 2. Income Group 3. Household Size Group 4. Children Count Group	34	Gaussian Mixture Model	i) number of components = 4 ii) covariance type = full	Yes	0.35	1.01	468.16	-
	35	Self Organizing Maps	i) sigma = 1.0 ii) learning rate = 0.5 iii) k = 3		0.36	0.92	476.38	

Table 7-2 (continued). Detailed information on validation measures, hyperparameters and usage of PCA for 35 clustering algorithms involved in the model selection

Variables	Model No	Clustering Algorithms	Number of Clusters	Cluster	Proportion	Cluster Proportion Description
RFM Variables 1. Recency 2. Frequency 3. Monetary	1	K-Means	4	1	0.282	The clusters are mildly imbalanced.
				2	0.230	
				3	0.243	
				4	0.245	
	2	Hierarchical Clustering (Ward Linkage)	2	1	0.622	The clusters are mildly imbalanced. However, 2 clusters are too simple to capture cluster structure and identify more subtle behaviours.
				2	0.378	
	3	Hierarchical Clustering (Average Linkage)	3	1	0.819	The clusters are moderately imbalanced.
				2	0.010	
				3	0.171	
	4	Hierarchical Clustering (Complete Linkage)	2	1	0.874	
				2	0.126	
	5	DBSCAN	3	0	0.087	The clusters are midly imbalanced. However, 2 clusters are too simple to capture cluster structure and identify more subtle behaviours. Cluster 0 is ignored in the cluster balance analysis because it represents noise.
				1	0.695	
				2	0.217	
	6	HDBSCAN	3	0	0.142	The clusters are moderately imbalanced. Cluster 0 is ignored in the cluster balance analysis because it represents noise.
				1	0.662	
				2	0.196	
Demographic Variables 1. Age Group 2. Income Group 3. Household Size Group 4. Children Count Group	7	K-Means	4	1	0.498	The clusters are moderately imbalanced.
				2	0.185	
				3	0.160	
				4	0.157	
	8	Hierarchical Clustering (Ward Linkage)	4	1	0.536	
				2	0.184	
				3	0.137	
				4	0.144	
	9	Hierarchical Clustering (Average Linkage)	4	1	0.140	The clusters are severely imbalanced.
				2	0.157	
				3	0.699	
				4	0.004	
	10	Hierarchical Clustering (Complete Linkage)	2	1	0.649	The clusters are mildly imbalanced. However, 2 clusters are too simple to capture cluster structure and identify more subtle behaviours.
				2	0.351	
	11	DBSCAN	17	0	0.034	i) The clusters are severely imbalanced, and 16 clusters are too granular for interpretation. ii) Cluster 0 is ignored in the cluster balance analysis because it represents noise.
				1	0.084	
				2	0.263	
				3	0.031	
				4	0.025	
				5	0.145	
				6	0.100	
				7	0.034	
				8	0.026	
				9	0.039	
				10	0.062	
				11	0.036	
				12	0.049	
				13	0.020	
				14	0.021	
				15	0.022	
				16	0.009	
	12	HDBSCAN	19	0	0.262	i) The clusters are severely imbalanced, 14 clusters are too granular for interpretation, and cluster 0 has high number of noise. ii) Cluster 0 is ignored in the cluster balance analysis because it represents noise
				1	0.021	
				2	0.050	
				3	0.079	
				4	0.031	
				5	0.006	
				6	0.025	
				7	0.142	
				8	0.022	
				9	0.095	
				10	0.039	
				11	0.020	
				12	0.024	
				13	0.024	
				14	0.034	
				15	0.024	
				16	0.021	
				17	0.047	
				18	0.034	

Table 7-3. Detailed information on the number of clusters, cluster proportions and cluster proportion descriptions for 35 clustering algorithms involved in the model selection process

Variables	Model No	Clustering Algorithms	Number of Clusters	Cluster	Proportion	Cluster Proportion Description
Behavioral Variables 1. Morning Proportion 2. Afternoon Proportion 2. Evening Proportion 4. Night Proportion 5. Midnight Proportion 6. Retail Discount Amount 7. Retail Discount Usage Proportion 8. Retail Discount Non-Usage Proportion 9. Coupon Discount Amount 10. Coupon Discount Usage Proportion 11. Coupon Discount Non-Usage Proportion 12. Campaign Non-Reach Proportion 13. Campaign Reach Proportion (16 Coupons) 14. Campaign Reach Proportion (All Coupons)	13	K-Means	2	1	0.111	The clusters are moderately imbalanced.
				2	0.889	
	14	Hierarchical Clustering (Ward Linkage)	2	1	0.900	
				2	0.100	
	15	Hierarchical Clustering (Average Linkage)	2	1	0.015	
				2	0.985	
	16	Hierarchical Clustering (Complete Linkage)	2	1	0.985	
				2	0.015	
	17	DBSCAN	3	0	0.089	The clusters are moderately imbalanced. Cluster 0 is ignored in the cluster balance analysis because it represents noise.
				1	0.089	
				2	0.823	
	18	HDBSCAN	3	0	0.095	
				1	0.084	
				2	0.821	
Product Preference Variables 1. National Brand Proportion 2. Private Brand Proportion 3. General Grocery Proportion 4. Health Personal CareProportion 5. Produce Proportion 6. Meat Proportion 7. Deli & Bakery Proportion 8. Miscellaneous & Services Proportion 9. Seafood Proportion 10. Home & Lifestyle Proportion 11. Entertainment & Leisure Proportion 12. Alcohol Proportion	19	K-Means	2	1	0.854	The clusters are moderately imbalanced.
				2	0.146	
	20	Hierarchical Clustering (Ward Linkage)	2	1	0.861	
				2	0.139	
	21	Hierarchical Clustering (Average Linkage)	2	1	0.996	The clusters are severely imbalanced.
				2	0.004	
	22	Hierarchical Clustering (Complete Linkage)	2	1	0.996	
				2	0.004	
	23	DBSCAN	3	0	0.035	The clusters are moderately imbalanced. Cluster 0 is ignored in the cluster balance analysis because it represents noise.
				1	0.126	
				2	0.839	
	24	HDBSCAN	3	0	0.026	
				1	0.135	
				2	0.839	
RFM, Demographic, Behavioral & Product Preference Variables 1. Recency 2. Frequency 3. Monetary 4. Age Group 5. Income Group 6. Household Size Group 7. Children Count Group 8. Morning Proportion 9. Afternoon Proportion 10. Evening Proportion 11. Night Proportion 12. Midnight Proportion 13. Retail Discount Amount 14. Retail Discount Usage Proportion 15. Retail Discount Non-Usage Proportion 16. Coupon Discount Amount 17. Coupon Discount Usage Proportion 18. Coupon Discount Non-Usage Proportion 19. Campaign Non-Reach Proportion 20. Campaign Reach Proportion (16 Coupons) 21. Campaign Reach Proportion (All Coupons) 22. National Brand Proportion 23. Private Brand Proportion 24. General Grocery Proportion 25. Health Personal CareProportion 26. Produce Proportion 27. Meat Proportion 28. Deli & Bakery Proportion 29. Miscellaneous & Services Proportion 30. Seafood Proportion 31. Home & Lifestyle Proportion 32. Entertainment & Leisure Proportion 33. Alcohol Proportion	25	K-Means	4	1	0.110	The clusters are moderately imbalanced.
				2	0.119	
				3	0.191	
				4	0.581	

Table 7-4 (continued). Detailed information on the number of clusters, cluster proportions and cluster proportion descriptions for 35 clustering algorithms involved in the model selection process

Variables	Model No	Clustering Algorithms	Number of Clusters	Cluster	Proportion	Cluster Proportion Description
RFM, Demographic & Behavioral Variables 1. Recency 2. Frequency 3. Monetary 4. Age Group 5. Household Size Group 6. Children Count Group 7. Morning Proportion 8. Afternoon Proportion 9. Evening Proportion 10. Night Proportion	26	K-Means	4	1	0.109	The clusters are moderately imbalanced.
				2	0.350	
				3	0.152	
				4	0.390	
RFM & Demographic Segmentations Variables 1. Recency 2. Frequency 3. Monetary 4. Age Group 5. Household Size Group 6. Children Count Group 7. Morning Proportion 8. Afternoon Proportion 9. Evening Proportion 10. Night Proportion	27	K-Means	4	1	0.160	The clusters are moderately imbalanced.
				2	0.321	
				3	0.275	
				4	0.245	
RFM & Demographic Variables 1. Recency 2. Frequency 3. Monetary 4. Age Group	28	K-Means	4	1	0.220	The clusters are moderately imbalanced.
				2	0.154	
				3	0.292	
				4	0.335	
RFM & Demographic Variables 1. Recency 2. Frequency 3. Monetary 4. Household Size Group 5. Children Count Group	29	K-Means	3	1	0.422	The clusters are moderately imbalanced.
				2	0.161	
				3	0.417	
RFM & Behavioural Variables 1. Recency 2. Frequency 3. Monetary 4. Morning Proportion 5. Afternoon Proportion 6. Evening Proportion 7. Night Proportion	30	K-Means	3	1	0.381	The clusters are moderately imbalanced.
				2	0.508	
				3	0.111	
Demographic & Behavioural Variables 1. Age Group 2. Household Size Group 3. Children Count Group 4. Morning Proportion 5. Afternoon Proportion 6. Evening Proportion 7. Night Proportion	31	K-Means	4	1	0.109	The clusters are moderately imbalanced.
				2	0.350	
				3	0.390	
				4	0.152	
RFM Variables 1. Recency 2. Frequency 3. Monetary	32	Gaussian Mixture Model	2	1	0.567	The clusters are well-balanced.
				2	0.433	
	33	Self Organizing Maps	4	1	0.233	The clusters are mildly imbalanced.
				2	0.256	
				3	0.199	
				4	0.312	
Demographic Variables 1. Age Group 2. Income Group 3. Household Size Group 4. Children Count Group	34	Gaussian Mixture Model	4	1	0.171	The clusters are moderately imbalanced.
				2	0.312	
				3	0.371	
				4	0.146	
	35	Self Organizing Maps	3	1	0.226	The clusters are moderately imbalanced.
				2	0.610	
				3	0.164	

Table 7-5 (continued). Detailed information on the number of clusters, cluster proportions and cluster proportion descriptions for 35 clustering algorithms involved in the model selection process.